Volatility Driven Distortions in Conditional Correlation

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January 1, 2026

1 Introduction

A well documented phenomenon is time-varying correlation between financial asset returns, often coinciding with periods of market tumult. A natural question to ask is whether the observed change in correlation reflects a change in the underlying joint distribution of returns, or is a statistical artifact. Boyer et al (1997)[2] and Loretan and English (2000)[8] argue that correlation changes observed over the 1990s are entirely consistent with a fixed unconditional correlation but changing conditional variance. Specifically, they argue that under joint normality, conditioning on events which inflate or deflate one asset's variance relative to population variance distorts subsample correlation, and thus observed correlation changes follow from the nature of conditional subsets, rather than a structural change.

I extend the conditional correlation framework in several regards. First, I provide an alternate proof of the key theorem underpinning both papers, which eliminates the explicit reliance on bivariate normality. Second, I derive an inversion formula, which expresses unconditional correlation as a function of observed conditional correlation and conditional variance. Third, I analyze the sampling variability of conditional variance and conditional correlation estimators by deriving closed-form expressions for conditional moments of tail and interior events. I further establish variance and kurtosis properties, and I use the delta method to quantify finite sample uncertainty. To complement our theoretical work and test accuracy and stability of derived results, I perform Monte Carlo experiments over a range of correlation levels and event types. Finally, I apply the Loretan-English framework to four decades of equity-rate and FX-equity data to assess whether volatility-induced effects can account for the pronounced time variation observed in realized correlations. Using up to date time series, I find that volatility conditioning doesn't explain enough of the observed time-varying correlation, thus reflecting genuine changes in the return process.

2 Literature Review

The central insight in Loretan and English and Boyer et al is that correlations measured during periods of market stress may increase mechanically as a consequence of volatility dynamics, rather than reflecting genuine changes in economic dependence. Given how pervasive rolling window correlations estimates are, and how often practitioners argue that changing correlations indicate a structural break, and despite the simplicity and explanatory power of the LE theorem, only a handful of papers cite Loretan and English or Boyer et al. For example, Hafner et al. (2006)[7]

motivate their semi-parametric conditional correlation framework by this identification problem, allowing correlations to evolve flexibly with volatility states instead of assuming structural correlation breakdowns. Similarly, Campbell et al. (2008)[4] design empirical tests to disentangle true dependence shifts from artifacts induced by fat tails and conditional heteroskedasticity, citing the Loretan-English argument as a key caution against naive interpretations of crisis-period correlation spikes. Because the Loretan–English contribution fundamentally demonstrates an identification problem, but doesn't solve it, the LE theorem appears to have been absorbed and used as a baseline assumption in subsequent papers (e.g. more elaborate multivariate GARCH and distribution-based approaches), rather than extended specifically.

3 Theoretical Extension

3.1 Alternate Proof

Both papers are predicated on the following theorem.

Theorem 1. *Consider a pair of bivariate normal random variables X and Y with variances σ_x^2 and σ_y^2, respectively, and covariance σ_{xy}. Let $\rho = \sigma_{xy}/(\sigma_x\sigma_y)$ denote the unconditional correlation. Then, for any event $A \subset \mathbb{R}$ with $0 < P(A) < 1$, the conditional correlation of X and Y given $X \in A$ is*

$$\rho_A = \rho \left[\rho^2 + (1-\rho^2) \frac{Var(X)}{Var(X \mid X \in A)} \right]^{-1/2}. \tag{1}$$

I plot bivariate normal draws from $X, Y \sim N(0,1)$, with $\rho = -0.5, 0.0, 0.5$, and color code observations corresponding to $A = \{(x,y) : |x| > 1.96\}$ (Figure 1) to provide geometric intuition for (1). I prove Theorem 1 using a minimal regression setup. Let the event $A = \{X \in \mathcal{A}\}$ occur

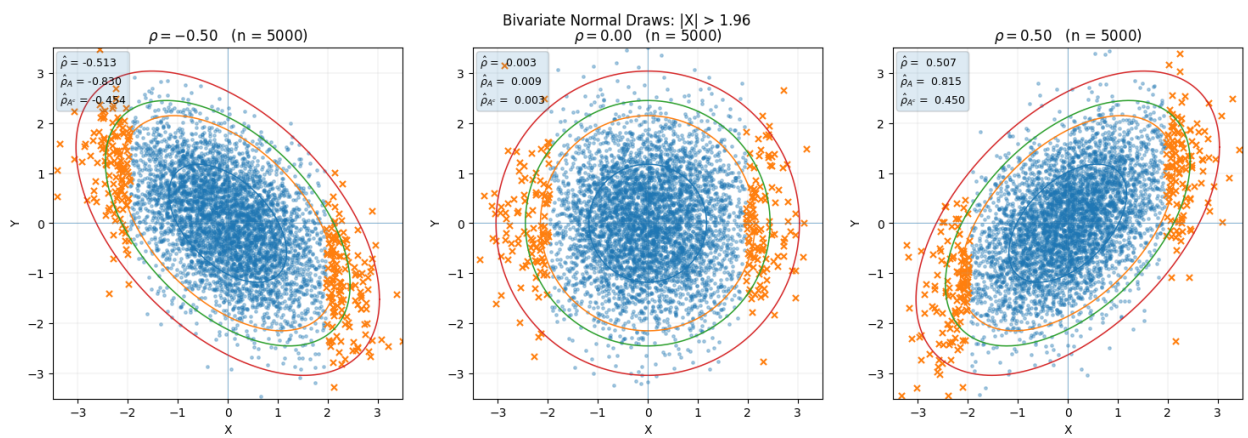


Figure 1: Scatterplots of $n = 5000$ draws from $(X,Y) \sim N(0,1)$ with correlation $\rho = -0.50, 0.0, 0.50$. Orange points correspond to $|X| > 1.96$. Confidence ellipses plotted at 50% (blue), 90% (orange), 95% (green), and 99% (red).

with $P(A) > 0$ and let $\mathrm{Var}_A(\cdot)$, $\mathrm{Cov}_A(\cdot,\cdot)$ denote variance and covariance conditional on A. Assume

$$Y \;=\; \beta X + U, \qquad \beta \;=\; \rho\,\frac{\sigma_y}{\sigma_x}, \qquad U \perp X, \qquad \mathrm{Var}(U) = \sigma_y^2(1-\rho^2).$$

Thus,

$$\mathbb{E}[Y \mid X] = \beta X, \qquad \mathrm{Var}(Y \mid X) = \sigma_y^2(1-\rho^2).$$

The conditional correlation is

$$\rho_A = \frac{\mathrm{Cov}_A(X,Y)}{\sqrt{\mathrm{Var}_A(X)}\,\sqrt{\mathrm{Var}_A(Y)}}.$$

The components are as follows. First, the conditional covariance:

$$\begin{aligned}\mathrm{Cov}_A(X,Y) &= \mathrm{Cov}_A(X, \beta X + U)\\ &= \beta \mathrm{Var}_A(X) + \mathrm{Cov}_A(X,U)\\ &= \beta \mathrm{Var}_A(X),\end{aligned}$$

using $U \perp X$. Second, the conditional variance of y. By the law of total variance,

$$\begin{aligned}\mathrm{Var}_A(Y) &= \mathbb{E}_A\big[\mathrm{Var}(Y \mid X)\big] + \mathrm{Var}_A\big(\mathbb{E}[Y \mid X]\big)\\ &= \sigma_y^2(1-\rho^2) + \beta^2 \mathrm{Var}_A(X).\end{aligned}$$

The conditional correlation is thus

$$\begin{aligned}\rho_A &= \frac{\beta \mathrm{Var}_A(X)}{\sqrt{\mathrm{Var}_A(X)}\sqrt{\sigma_y^2(1-\rho^2) + \beta^2 \mathrm{Var}_A(X)}}\\ &= \rho\,\frac{\sigma_y}{\sigma_x}\,\frac{\sqrt{\mathrm{Var}_A(X)}}{\sqrt{\sigma_y^2(1-\rho^2) + \rho^2 \frac{\sigma_y^2}{\sigma_x^2}\mathrm{Var}_A(X)}}\\ &= \rho\left[\frac{\rho^2 \mathrm{Var}_A(X) + \sigma_x^2(1-\rho^2)}{\mathrm{Var}_A(X)}\right]^{-1/2}.\end{aligned}$$

Finally,

$$\boxed{\rho_A = \rho\left[\rho^2 + (1-\rho^2)\,\frac{\sigma_x^2}{\mathrm{Var}_A(X)}\right]^{-1/2}.}$$

It follows that if $\mathrm{Var}(X \mid A) > \mathrm{Var}(X)$, then $|\rho_A| > |\rho|$. If $\mathrm{Var}(X \mid A) < \mathrm{Var}(X)$, then $|\rho_A| < |\rho|$. Conditioning on a higher (lower) variance subset raises (lowers) conditional correlation relative to population correlation. Without loss of generality, standardize X and Y such that $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$, and let $k = \mathrm{Var}(X)/\mathrm{Var}(X \mid A)$. Thus

$$\rho_A = \frac{\rho}{\sqrt{\rho^2 + (1-\rho^2)\,k}}.$$

Note that if $\rho = 0$, then $\rho_A = 0$ for every A, if $|\rho| = 1$, then $\rho_A = \rho$ for every A, and if $k = 1$ (no variance distortion), then $\rho_A = \rho$. Figure 11 illustrates the dependence of conditional correlation on conditional variance. For fixed population ρ, conditioning on low-variance subsamples drives ρ_A toward zero, whereas conditioning on high-variance subsamples drives ρ_A toward ± 1, consistent with the limiting behavior of (1). Intuitively, conditioning on a very small slice of X pushes conditional variance to 0 and destroys co-movement between X and Y, thus pushing conditional correlation towards 0. Conditioning on extreme X observations raises conditional variance and draws from the tails of the ellipse, which thin out to nearly a straight line for more extreme X.

3.2 Inversion

Theorem 1 can be inverted to express the population correlation ρ as a function of observed conditional correlation ρ_A and variance ratio k:

$$\rho = \operatorname{sign}(\rho_A)\sqrt{\frac{\rho_A^2\, k}{1-\rho_A^2(1-k)}}, \qquad k = \frac{\operatorname{Var}(X)}{\operatorname{Var}(X \mid A)}.$$

which is well-defined when $1 - \rho_A^2(1-k) > 0$. Figure 11 shows contour plots of implied correlation for a subsample ρ_A and variance ratio k grid, providing a quick reference to identify combinations of subsample correlations and variance ratios associated with the same population correlation. Note, I demonstrate algebraic invertibility, but statistical recovery of the population correlation from sample data is a separate question.

4 Monte Carlo Evaluation

4.1 Conditional Correlation and Inversion Stability

I conduct a Monte Carlo experiment using bivariate normal returns with unconditional correlations $\rho \in \{0.5, 0.95\}$. Our objectives are twofold. First, to validate the conditional-correlation prediction ρ_A under realistic sample sizes and empirically defined conditioning events. Second, to assess inversion stability, meaning whether the underlying unconditional correlation ρ can reliably be recovered from observed conditional correlations. I draw i.i.d. samples (X_i, Y_i) from $N_2(0, \Sigma)$ with $\operatorname{Var}(X) = \operatorname{Var}(Y) = 1$ and $\operatorname{Corr}(X, Y) = \rho$ and form subsamples using

1. Decile-delimited intervals of X, defined by the empirical deciles of the simulated X sample
2. Absolute-tail events $A = \{|X| \geq q\}$ for high empirical quantiles q of $|X|$

For each event A I compute the empirical conditional correlation $\hat{\rho}_A$, the Loretan–English prediction ρ_A obtained from equation (1), the variance ratio $k = \operatorname{Var}(X)/\operatorname{Var}(X \mid A)$, and the inverted estimate ρ_{inv} obtained by solving the same mapping for ρ. Across both designs, the LE predictions track the simulated conditional correlations quite well. Interior deciles, which have small $\operatorname{Var}(X \mid$

A), exhibit dampened conditional correlations relative to the full sample, while tail events, with large $\mathrm{Var}(X \mid A)$, exhibit inflated conditional correlations, matching the theoretical dependence of ρ_A on the conditional variance. Remarkably, ρ_{inv}'s recovery of the true unconditional correlation is highly accurate in both moderate and high-ρ settings for tail events[1], despite sometimes small subsamples and Monte Carlo noise[2]. The LE mapping is numerically stable and unconditional correlations can, in principle, be recovered from conditional subsamples.

4.2 Grid Evaluation

The preceding simulation assessed accuracy and stability of the Loretan-English conditional correlation formula and inversion at specific ρ values for tail and interior decile events. I conduct a large-scale simulation, fixing the event but varying ρ over a grid, to do the following. First, evaluate whether the LE formula remains accurate uniformly over $\rho \in [-1, 1]$, rather than at isolated benchmark values. Second, examine the sampling variability and highlight regions where inference and inversion becomes unstable (especially near $\rho = 0$). I proceed as follows.

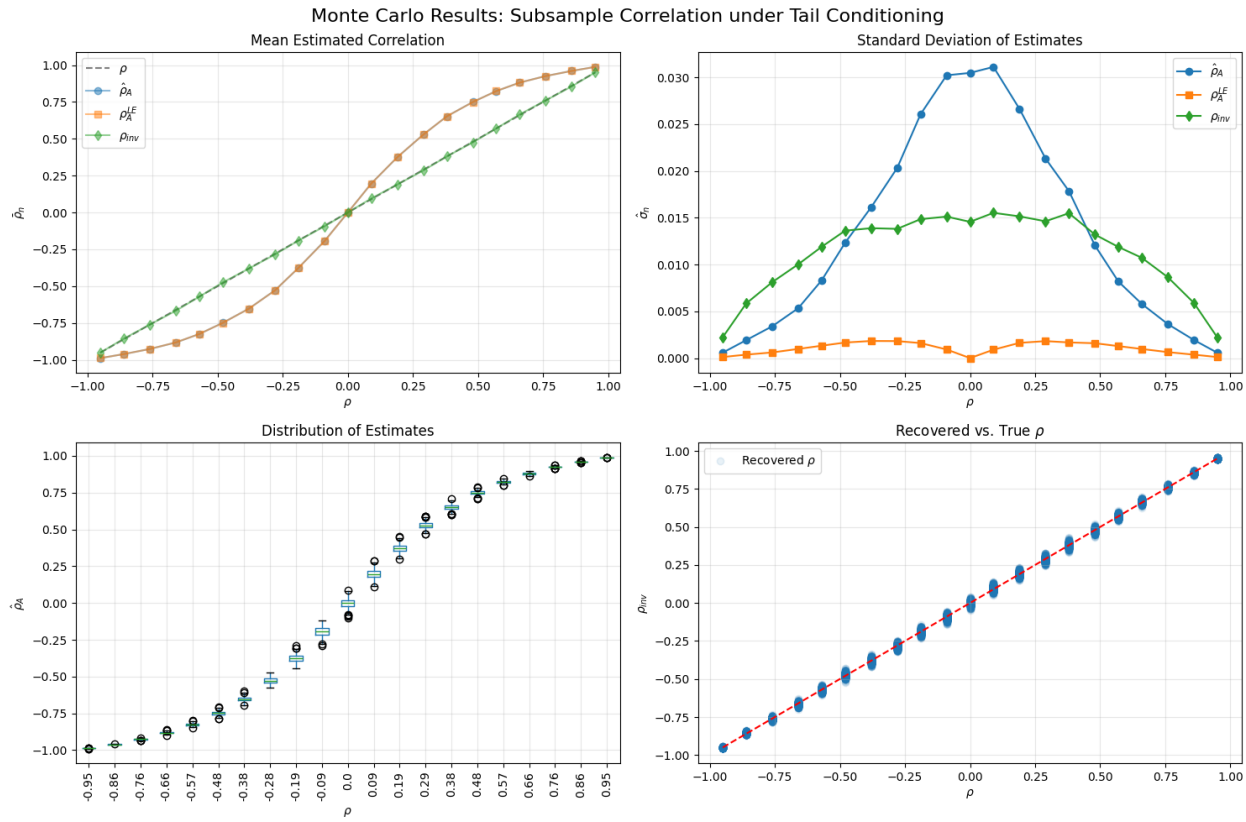
For 21 ρ values in a grid from -0.95 to 0.95 (inclusive), I generate $R = 500$ independent datasets of size $n = 10{,}000$, drawing $\{(X_i, Y_i)\}_{i=1}^{n}$ i.i.d. from a bivariate normal distribution with unit variances and covariance ρ. Within each dataset, I condition on the tail event

$$A = \{\, |X| \geq q \,\}, \qquad q = \text{empirical } 90^{\text{th}} \text{ percentile of } |X|.$$

I use a data-dependent threshold, despite it introducing additional Monte Carlo noise (both the membership of A and the conditional variance vary across simulations even when ρ is fixed), to assess performance of the theorem as it is commonly used. Further mirroring practice, and the approach of Boyer et al and Loretan and English, I treat the complete sample's correlation and variance as fixed population inputs. For each subsample, I compute the empirical conditional correlation $\hat{\rho}_A$, the LE prediction ρ_A^{LE}, and the inverted estimate ρ_{inv}. Repeating this procedure over the ρ grid yields distributions of empirical, theoretical, and inverted correlations, summarized in the figure below.

[1] Inversion performs worse for interior deciles when $\rho = 0.5$ because empirical decile boundaries generate extremely small conditional variances. Inversion is non-linear, thus the sampling variability in conditional variances plus small Monte Carlo fluctuations produce disproportionately large distortions in the inverted correlation.

[2] Several design features contribute to Monte Carlo noise. First, both the decile boundaries and tail thresholds are defined empirically, so the conditioning events and their associated conditional variances are random. Second, the most extreme tail sets contain relatively few observations (down to $N_{\mathrm{eff}} = 100$ in the $q = 0.99$ case), which amplifies typical sampling variability. Third, the LE predictions and inversions use estimated conditional variances rather than population values, so noise in $\widehat{\mathrm{Var}}(X \mid A)$ propagates into ρ_A and ρ_{inv}. I identify modest discrepancies between empirical and LE quantities in the thin tails, and attribute those to the above sources. See Appendix A for results.



I observe the following:

1. Across the ρ grid, average empirical subsample correlation (blue) matches the LE theoretical prediction (orange) almost perfectly. Inversion (green) successfully reconstructs the true correlation (black) (See top left).

2. The empirical subsample correlation $\hat{\rho}_A$ exhibits considerable variance, maximally near $\rho = 0$ and diminishing as $|\rho| \to 1$. This occurs because when the unconditional correlation is weak, the signs of (X, Y) in the tail region are driven more by sampling noise than by underlying dependence. The variance of LE predictions (orange) can be interpreted as a floor because the LE prediction is a deterministic function of (k, ρ). The inverted estimator ρ_{inv} has lower variance than $\hat{\rho}_A$, and displays the same instability near $\rho = 0$ where the denominator in the inversion becomes small (See top right).

3. The boxplots reveal skewness in the distribution of $\hat{\rho}_A$ (upward when $\rho > 0$ and downward when $\rho < 0$). This arises because conditioning on large $|X|$ disproportionately selects observations where Y shares the sign of X when $\rho > 0$, and the opposite when $\rho < 0$. Even when $\rho = 0$, the tail-subset inherits sign asymmetry from the realized sample, yielding asymmetric distributions of $\hat{\rho}_A$. Dispersion again peaks near $\rho = 0$, suggesting that inference in this region is relatively fragile (See bottom left).

4. The recovered vs. true scatterplot shows that ρ_{inv} lies tightly on the 45-degree line except near $\rho = 0$ (numerical instability) and near $|\rho| = 1$ (boundary compression). Practically, this suggests that inversion works reliably unless the true correlation is extremely small or extremely close to ± 1 (See bottom right).

For an extreme tail event, the LE theorem holds empirically over almost the entirety of the correlation range. Estimation is difficult when $\rho \approx 0$ and inversion is numerically sensitive when $\rho \approx 0$ and $|\rho| \approx 1$.

5 Computing Conditional Variance

After normalizing $\mathrm{Var}(X)$ and fixing unconditional correlation ρ, population conditional correlation ρ_A depends solely on $\mathrm{Var}(X \mid A)$. Boyer et al (1997) outlines an integral form for conditional variance when X is Gaussian, thus in this section I provide analytic expressions when A is a symmetric tail or interior event and detail simulation procedures to address insufficient sample size.

5.1 Symmetric Tail Event

Consider events of the form $A = \{|X| > c\}$, where $X \sim N(0,1)$, $c > 0$, and $P(A) > 0$. Using

$$\mathrm{Var}(X \mid A) = \mathbb{E}[X^2 \mid A] - \big(\mathbb{E}[X \mid A]\big)^2,$$

and noting that both the standard normal density and the event $\{|X| > c\}$ are symmetric, and thus the conditional distribution $X \mid |X| > c$ is symmetric around zero, it follows that

$$\mathbb{E}[X \mid |X| > c] = 0, \qquad \mathrm{Var}(X \mid |X| > c) = \mathbb{E}[X^2 \mid |X| > c].$$

For any random variable Y and event A,

$$\mathbb{E}[Y \mid A] = \frac{\mathbb{E}[Y\,\mathbf{1}_A]}{P(A)}.$$

Let $Y = X^2$ and $A = \{|X| > c\}$, thus

$$\mathbb{E}[X^2 \mid |X| > c] = \frac{\mathbb{E}[X^2\,\mathbf{1}_{\{|X|>c\}}]}{P(|X| > c)}.$$

I solve for the denominator first. By symmetry,

$$P(|X| > c) = 2\,P(X > c) = 2\big(1 - \Phi(c)\big) = 2\,\Phi(-c),$$

where ϕ and Φ denote the standard normal PDF and CDF, respectively. The event $\{|X| > c\}$ is a two-sided tail region, $(-\infty, -c] \cup [c, \infty)$, thus I solve for the numerator by first partitioning the integral,

$$\mathbb{E}[X^2] = \int_{-\infty}^{-c} x^2\phi(x)\,dx + \int_{-c}^{c} x^2\phi(x)\,dx + \int_{c}^{\infty} x^2\phi(x)\,dx = 1.$$

Thus

$$\begin{aligned}\mathbb{E}\big[X^2\mathbf{1}_{\{|X|>c\}}\big] &= \int_{-\infty}^{-c} x^2\phi(x)\,dx + \int_{c}^{\infty} x^2\phi(x)\,dx \\ &= 1 - \int_{-c}^{c} x^2\phi(x)\,dx.\end{aligned}$$

Because $x^2\phi(x)$ is even,

$$\int_{-c}^{c} x^2\phi(x)\,dx = 2\int_{0}^{c} x^2\phi(x)\,dx,$$

and it can be shown using integration by parts[6][1] that $\int_0^c x^2\phi(x)\,dx$ has the closed form

$$2\int_{0}^{c} x^2\phi(x)\,dx = -2c\,\phi(c) + 2\Phi(c) - 1,$$

thus

$$\mathbb{E}\big[X^2\mathbf{1}_{\{|X|>c\}}\big] = 1 - \big(-2c\phi(c) + 2\Phi(c) - 1\big) = 2 + 2c\phi(c) - 2\Phi(c).$$

Combine the numerator and denominator, and using $\Phi(-c) = 1 - \Phi(c)$,

$$\begin{aligned}\mathbb{E}[X^2 \mid |X| > c] &= \frac{2 + 2c\phi(c) - 2\Phi(c)}{2\big(1 - \Phi(c)\big)} \\ &= \frac{1 + c\phi(c) - \Phi(c)}{1 - \Phi(c)} \\ &= 1 + \frac{c\,\phi(c)}{\Phi(-c)}.\end{aligned}$$

Finally (recall the conditional mean is zero), note tail variance exceeds unconditional unit variance:

$$\boxed{\mathrm{Var}(X \mid |X| > c) = 1 + \frac{c\,\phi(c)}{\Phi(-c)}.}$$

thus $k < 1$ and $\mid \rho_a \mid > \mid \rho \mid$.

5.2 Interior Events

Consider symmetric interior events such as $A = \{|X| < c\}$, or more generally $A = \{a < X < b\}$. Unlike conditioning on tail events, which inflate dispersion, conditioning on interior events reduces conditional variance by concentrating mass near the center of the distribution. Fortunately, the conditional distribution for interior events is truncated normal, with readily available closed forms

for the conditional mean and variance. Let $X \sim N(\mu, \sigma^2)$, and consider the event

$$A = \{\, a < X < b \,\}, \qquad a < b.$$

and define standardized bounds

$$\alpha = \frac{a-\mu}{\sigma}, \qquad \beta = \frac{b-\mu}{\sigma}.$$

Following Burkardt (2023)[3], the conditional mean and variance of the truncated normal are

$$\mathbb{E}[X \mid a < X < b] = \mu + \sigma\, \frac{\phi(\alpha) - \phi(\beta)}{\Phi(\beta) - \Phi(\alpha)},$$

$$\mathrm{Var}(X \mid a < X < b) = \sigma^2 \left(1 - \frac{\beta\phi(\beta) - \alpha\phi(\alpha)}{\Phi(\beta) - \Phi(\alpha)} - \left(\frac{\phi(\alpha) - \phi(\beta)}{\Phi(\beta) - \Phi(\alpha)} \right)^2 \right).$$

Now let $X \sim N(0,1)$, thus $\alpha = a$, $\beta = b$. Conditional mean and variance simplify to

$$\mathbb{E}[X \mid a < X < b] = \frac{\phi(a) - \phi(b)}{\Phi(b) - \Phi(a)},$$

$$\mathrm{Var}(X \mid a < X < b) = 1 - \frac{b\phi(b) - a\phi(a)}{\Phi(b) - \Phi(a)} - \left(\frac{\phi(a) - \phi(b)}{\Phi(b) - \Phi(a)} \right)^2.$$

Now let $a = -c$ and $b = c$. Note ϕ is even and $\Phi(-c) = 1 - \Phi(c)$, thus

$$\phi(-c) = \phi(c), \qquad \phi(a) - \phi(b) = 0, \qquad \Phi(b) - \Phi(a) = 2\Phi(c) - 1.$$

The conditional mean, $\mathbb{E}[X \mid |X| < c]$ is zero, thus the conditional variance simplifies to

$$\mathrm{Var}(X \mid |X| < c) = 1 - \frac{c\,\phi(c) - (-c)\phi(c)}{2\Phi(c) - 1}.$$

Finally[3],

$$\boxed{\mathrm{Var}(X \mid |X| < c) = 1 - \frac{2c\,\phi(c)}{2\Phi(c) - 1}}.$$

I verify these analytical results using simulation C.

5.3 Monte Carlo Estimation of Var$(X \mid A)$

I distinguish between two challenges for Monte Carlo estimation of conditional variance. The first is that X is non-Gaussian, or A is complicated (e.g. unions of intervals, a non-linear function of X), such that a closed form expression for $\mathrm{Var}(X \mid A)$ may not exist or may be computationally

[3] Note that for $c > 0$, $2\Phi(c) - 1 > 0$, thus $\mathrm{Var}(X \mid |X| < c) < 1$, as expected for an interior truncation. Thus $k > 1$ and $\mid \rho_a \mid < \mid \rho \mid$.

intractable. The second is that even when the conditional distribution $X \mid A$ is known in closed form (as with truncated normals), naive unconditional sampling may produce an insufficient effective sample size. The former is beyond the scope of this paper. I address the latter.

Let $X \sim p$ and draw $Z_1, \ldots, Z_n$ i.i.d. from p, keeping the subsample $\{Z_i : Z_i \in A\}$. Define N_{eff} as the number of such samples that survive conditioning. The natural Monte Carlo estimator is

$$\widehat{\text{Var}}(X \mid A) = \frac{1}{N_{\text{eff}} - 1} \sum_{i: Z_i \in A} \left(Z_i - \overline{Z}_A\right)^2, \qquad \overline{Z}_A = \frac{1}{N_{\text{eff}}} \sum_{i: Z_i \in A} Z_i.$$

Note that N_{eff} varies per sample, and thus is random with $\mathbb{E}[N_{\text{eff}}] = nP(A)$[4]. For an event A with low $P(A)$, a small fraction of unconditional draws fall inside the event. Consequently, the conditional variance estimator is inherently noisy when the event A is rare as the estimate is dominated by Monte Carlo error.[5]. I describe three strategies for reducing Monte Carlo error.

(i) Multiple Runs: Repeat the unconditional simulation R times, computing $\widehat{\text{Var}}^{(1)}(X \mid A) \ldots \widehat{\text{Var}}^{(r)}(X \mid A)$. Averaging may stabilize the estimate and conveniently gives a measure of Monte Carlo error,

$$\widehat{\text{Var}}_{\text{MC}}(X \mid A) = \frac{1}{R} \sum_{r=1}^{R} \widehat{\text{Var}}^{(r)}(X \mid A), \qquad \text{Error}_{\text{MC}} = \sqrt{\frac{1}{R-1} \sum_{r=1}^{R} \left(\widehat{\text{Var}}^{(r)}(X \mid A) - \widehat{\text{Var}}_{\text{MC}}(X \mid A) \right)^2}.$$

Nonetheless, each run may still contain very few points in event A.

(ii) Rejection Sampling: Draw repeatedly from the unconditional distribution and accept a sample if it lies in A. Accepted values are i.i.d from the conditional distribution $X|A$, thus their sample variance gives an estimate of $\text{Var}(X \mid A)$. The upside is this approach decouples Monte Carlo accuracy from the magnitude of $P(A)$, because if the event is rare, the algorithm simply produces more rejections before collecting a prescribed target number M of accepted draws. The downside is computational (Robert and Casella 2004)[9]. On average, one must generate $1/P(A)$ proposals for each accepted draw.[6]

(iii) Direct Sampling: When the conditional distribution $X|A$ has a standard parametric form, e.g. the Gaussian setting where $X \mid A$ is truncated normal, simulate directly from $X \mid A$ using software (e.g. `scipy.stats.truncnorm`). This eliminates both the randomness of N_{eff} and the inefficiency of rejection sampling. The chosen conditional sample size dictates MC precision.

[4]Formally, $N_{\text{eff}} = \sum_{i=1}^{n} \mathbf{1}\{Z_i \in A\}$. As draws are i.i.d, the expected value is n times the expected value of the indicator function, which is just $P(A)$.

[5]For example, if $X \sim N(0,1)$ and $A = \{|X| > 1\}$, then $P(A) \approx 0.317$. For more extreme tail sets, $P(A)$ rapidly approaches zero, and the number of surviving samples becomes negligible relative to n.

[6]See Appendix B for a discussion of computational efficiency.

Despite the conceptual simplicity of rejection sampling, for Monte Carlo simulations I use (i) and (iii) where applicable, and otherwise draw a large enough n for sufficient N_{eff}.

6 Sampling Variability of the Conditional Variance Estimator

Applying the conditional correlation theorem requires an estimate of conditional variance. Even with sufficient N_{eff}, the conditional variance estimator will exhibit some sampling variability. In this section, I derive a measure of relative standard deviation to inform the choice of unconditional sample size for simulations. Let $\{X_i\}_{i=1}^{N_{\text{eff}}}$ be i.i.d. draws from the conditional distribution $X \mid A$, assumed to be mean 0, with variance $\sigma_A^2 = \text{Var}(X \mid A)$, and fourth (central) moment $\mu_4 = \mathbb{E}[X^4 \mid A]$. Per Casella and Berger (2002)[5], the unbiased sample variance

$$S_A^2 = \frac{1}{N_{\text{eff}} - 1} \sum_{i=1}^{N_{\text{eff}}} (X_i - \bar{X}_A)^2, \qquad \bar{X}_A = \frac{1}{N_{\text{eff}}} \sum_{i=1}^{N_{\text{eff}}} X_i,$$

has sampling variance

$$\begin{aligned} \text{Var}(S_A^2) &= \frac{1}{N_{\text{eff}}} \left[\mu_4 - \left(\frac{N_{\text{eff}} - 3}{N_{\text{eff}} - 1} \right) \sigma_A^4 \right] \\ &= \frac{\sigma_A^4}{N_{\text{eff}}} \left(\kappa_A - \frac{N_{\text{eff}} - 3}{N_{\text{eff}} - 1} \right), \end{aligned}$$

where $\kappa_A = \mu_4 / \sigma_A^4$ is the kurtosis of $X \mid A$. Note that for large N_{eff},

$$\text{Var}(S_A^2) \approx \frac{(\kappa_A - 1)\sigma_A^4}{N_{\text{eff}}}, \qquad \text{SD}(S_A^2) \approx \sigma_A^2 \sqrt{\frac{\kappa_A - 1}{N_{\text{eff}}}},$$

which gives a simple rule of thumb. The relative standard deviation of the conditional variance estimator measures variability relative to the mean and can be written as

$$\text{RSD}_A = \frac{\text{SD}(S_A^2)}{\sigma_A^2} = \sqrt{\frac{\kappa_A - \frac{N_{\text{eff}} - 3}{N_{\text{eff}} - 1}}{N_{\text{eff}}}}.$$

Note that N_{eff} and κ_A completely determine the RSD. I use RSD_A to choose unconditional sample size n for Monte Carlo experiments and inform sampling variability in historical time series data based on realized N_{eff}. For example, setting $\text{RSD}_A = 0.01$ gives

$$N_{\text{eff}} \approx \frac{\kappa_A - 1}{0.01^2}.$$

with corresponding unconditional sample size $n \approx N_{\text{eff}}/P(A)$. For symmetric truncations of a standard normal $X \sim N(0, 1)$, I derive σ_A^2 and κ_A analytically for both interior events $A = \{|X| < c\}$ and tail events $A = \{|X| > c\}$, using truncated-normal formulas for the former and the law of total expectation for the latter. I perform Monte Carlo simulation to assess the performance of

the conditional variance estimator. Across a range of cutoffs c, sample sizes n, and both tail and interior events, once $N_{\text{eff}} \gtrsim 300$, the MC conditional variance estimator stabilizes to within 10^{-3} of the analytic value. When N_{eff} falls below 50, the estimates degrade substantially, particularly for tail events where higher kurtosis further amplifies variance. Our results suggest that using a fixed unconditional sample size of $n = 10{,}000$ suffices for accurate estimation of conditional variance[7].

7 Sampling Variability of the Conditional Correlation Estimator

In practice, empirical conditional correlations are computed from a single realization of N_{eff} data points that fall inside the conditioning event. Loretan and English and Boyer et al both treat the full-sample variance and correlation as fixed inputs for population unconditional $\text{Var}(X)$ and ρ, respectively, thus *sampling noise enters exclusively through the conditional variance estimator* S_A^2. Accordingly, the conditional variance estimator's finite-sample distribution controls the sampling variability of the estimator $\widehat{\rho}_A^{LE}$[8].

I outline a framework for the propagation of conditional-variance estimator noise into the conditional correlation estimator, and evaluate the framework using Monte Carlo simulationD.1. Normalize $\text{Var}(X)$ to 1 and let the LE estimator be a function of sample conditional variance S_A^2

$$\widehat{\rho}_A^{LE} = g(S_A^2), \qquad g(z) = \frac{\rho}{\sqrt{\rho^2 + (1-\rho^2)\,\dfrac{1}{z}}}.$$

Using the delta method, expand around the true conditional variance $z = \sigma_A^2$:

$$\text{Var}(\widehat{\rho}_A^{LE}) \approx \left(g'(\sigma_A^2)\right)^2 \text{Var}(S_A^2).$$

I first compute $g'(z)$ as follows:

$$\frac{d}{dz}\left[\rho^2 + (1-\rho^2)\,z^{-1}\right]^{-1/2} = -\frac{1}{2}\left[\rho^2 + (1-\rho^2)\,z^{-1}\right]^{-3/2} \cdot \left(-(1-\rho^2)\,z^{-2}\right)$$
$$g'(z) = \rho \cdot \frac{1}{2}(1-\rho^2)\,z^{-2}\left[\rho^2 + (1-\rho^2)\,z^{-1}\right]^{-3/2}.$$

[7]For interior events, $P(|X| < c)$ is large, so $N_{\text{eff}} \approx 3{,}800,\ 6{,}800, 9{,}500$ for $c = 0.5,\ 1,$ and 2, respectively. With thousands of surviving samples, the MC variance estimate is extremely close to the analytic result. Even for the most extreme tail event I consider ($c = 2$, $P(A) \approx 0.0455$), $n = 10{,}000$ yields $N_{\text{eff}} \approx 455$, which is large enough that the MC estimate deviates from the truth by only a few hundredths. When desired, one can directly target a required effective sample size N_{eff}^* by selecting $n \approx N_{\text{eff}}^*/P(A)$. For example, attaining $N_{\text{eff}}^* \approx 500$ in the $c = 2$ tail case requires only $n \approx 500/0.0455 \approx 11{,}000$. Full derivations of the conditional moments, the computation of κ_A, and the simulation results are presented in Appendix C.

[8]Even with accurate unconditional inputs ρ and $\text{Var}(X)$, the conditional variance alone induces notable sampling uncertainty. Consider $X \sim N(0,1)$ and $A = \{|X| > 2\}$, thus $P(|X| > 2) = 0.0455$ and $\sigma_A^2 \approx 5.746$, $\kappa_A \approx 1.10$. If only $N_{\text{eff}} = 60$ points lie in the tail (e.g. one quarter of trading days), then $\text{RSD}_A \approx \sqrt{0.10/60} \approx 4.0\%$.

Evaluating at $z = \sigma_A^2$ gives

$$g'(\sigma_A^2) = \frac{\rho(1-\rho^2)}{2\,\sigma_A^4}\left[\rho^2 + \frac{1-\rho^2}{\sigma_A^2}\right]^{-3/2}.$$
$$\left(g'(\sigma_A^2)\right)^2 = \frac{\rho^2(1-\rho^2)^2}{4\,\sigma_A^8}\left[\rho^2 + \frac{1-\rho^2}{\sigma_A^2}\right]^{-3},$$

and combining with the exact expression for $\mathrm{Var}(S_A^2)$ yields the approximate sampling variance of the LE estimator:

$$\mathrm{Var}(\widehat{\rho}_A^{LE}) \approx \left[\frac{\rho^2(1-\rho^2)^2}{4\,\sigma_A^8}\left(\rho^2 + \frac{1-\rho^2}{\sigma_A^2}\right)^{-3}\right]\left[\frac{\sigma_A^4}{N_{\text{eff}}}\left(\kappa_A - \frac{N_{\text{eff}}-3}{N_{\text{eff}}-1}\right)\right]$$

Finally,

$$\boxed{\mathrm{Var}(\widehat{\rho}_A^{LE}) \approx \frac{\kappa_A - \frac{N_{\text{eff}}-3}{N_{\text{eff}}-1}}{N_{\text{eff}}} \cdot \frac{\rho^2(1-\rho^2)^2}{4\sigma_A^4}\left(\rho^2 + \frac{1-\rho^2}{\sigma_A^2}\right)^{-3}.}$$

with approximate standard deviation:

$$\boxed{\mathrm{SD}(\widehat{\rho}_A^{LE}) \approx \sqrt{\frac{\kappa_A - \frac{N_{\text{eff}}-3}{N_{\text{eff}}-1}}{N_{\text{eff}}}} \cdot \frac{|\rho|(1-\rho^2)}{2\sigma_A^2}\left(\rho^2 + \frac{1-\rho^2}{\sigma_A^2}\right)^{-3/2}.}$$

Note the following. First, $\mathrm{SD}(\widehat{\rho}_A^{LE}) \propto 1/\sqrt{N_{\text{eff}}}$, and thus scales inversely with $\sqrt{N_{\text{eff}}}$. Higher (lower) effective sample size reduces (increases) the standard deviation of the estimator's sampling distribution[9]. Second, the standard deviation is increasing in conditional kurtosis but the relationship with conditional variance is complicated by nonlinearity of $g(\cdot)$. Counterintuitively, *larger* conditional variance σ_A^2 *reduces* the sampling variability of $\widehat{\rho}_A^{LE}$. In very extreme tails (where σ_A^2 is large), the LE denominator $\rho^2 + (1-\rho^2)/\sigma_A^2$ is dominated by ρ^2, making $\widehat{\rho}_A^{LE}$ close to $\mathrm{sign}(\rho)$, and hence insensitive to moderate fluctuations in S_A^2. As the tail becomes more extreme, the LE estimator becomes more stable with respect to variance estimation noise, even though N_{eff} may be small[10]. Third, the dependence on tail kurtosis κ_A (increasing) implies that heavier-tailed distributions create greater variability in S_A^2 and therefore in $\widehat{\rho}_A^{LE}$. Empirically, this phenomenon is known as 'correlation spikes' in turbulent markets. Extreme-tail conditioning magnifies the influence of fourth-moment behavior, and heavy-tailed data amplify the sampling volatility of conditional correlation estimates.

[9] The square root means accuracy improves modestly with additional tail observations. For example, cutting the standard deviation in half requires roughly four times as many N_{eff}. Achieving $\mathrm{SD}(\widehat{\rho}_A^{LE}) \approx 0.001$ in the $c = 2$ tail event ($P(A) \approx 0.0455$) would require $N_{\text{eff}} \approx 2500$ tail points, corresponding to $2500/0.0455 \approx 55{,}000$ total observations, which is more than two centuries of daily data. The LE estimator faces inherent accuracy limits with rare tail events.

[10] To illustrate the effect, consider $X \sim N(0,1)$, $\rho = 0.5$, and fix $N_{\text{eff}} = 60$. For tail event $A = \{|X| > 2\}$, $\sigma_A^2 \approx 5.746$ and $\kappa_A \approx 1.096$. Substituting these gives $\mathrm{SD}(\widehat{\rho}_A^{LE}) \approx 0.0065$. For $A = \{|X| > 3\}$, $\sigma_A^2 \approx 10.84$ while $\kappa_A \approx 1.033$, giving $\mathrm{SD}(\widehat{\rho}_A^{LE}) \approx 0.0032$. See D.2 for a complete example.

8 Confidence Intervals for Conditional Correlation

Confidence intervals are a natural extension of the discussion on sampling variability. Thus far, we've considered the case where unconditional correlation ρ and unconditional variance $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$ are treated as known, and the only randomness comes from estimating the conditional variance $S_A^2 \approx \mathrm{Var}(X \mid A)$. Sampling variation in $\widehat{\rho}_A^{LE} = g(S_A^2)$ is therefore driven solely by the finite-sample behavior of S_A^2. However, for conditional Pearson correlations computed on N_{eff} observations that fall in event A, sampling variation reflects the entire scheme. Boyer et al and Loretan and English provide no substantive commentary (Boyer bootstraps 90% CIs), thus I compute confidence intervals using Fisher's z-transformation that reflect the totality of sampling variability and tighter confidence intervals reflecting only conditional variance.

8.1 Fisher's z-transformation

Let (X_i, Y_i), $i = 1, \ldots, N_{\text{eff}}$[11], be i.i.d. draws from a bivariate normal distribution with correlation ρ_A. The sample correlation

$$r_A = \widehat{\rho}_A = \frac{\sum_{i=1}^{N_{\text{eff}}} (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^{N_{\text{eff}}} (X_i - \bar{X})^2}\sqrt{\sum_{i=1}^{N_{\text{eff}}} (Y_i - \bar{Y})^2}}$$

is biased and bounded, with a skewed sampling distribution (especially for large $|r_A|$ or small N_{eff}). Fisher's z-transformation normalizes and stabilizes the variance of the correlation estimate (Welz 2020)[10]

$$z_A = \operatorname{arctanh}(r_A) = \frac{1}{2}\log\left(\frac{1 + r_A}{1 - r_A}\right).$$

Under bivariate normality,

$$z_A \approx N\left(\operatorname{arctanh}(\rho_A), \frac{1}{N_{\text{eff}} - 3}\right),$$

thus $\widehat{z}_A = \operatorname{arctanh}(r_A)$ is approximately normal with mean $\operatorname{arctanh}(\rho_A)$ and standard error $1/\sqrt{N_{\text{eff}} - 3}$ (Wicklin 2017)[11]. Construct a two-sided $(1 - \alpha)$ Fisher-z confidence interval for ρ_A as follows:

$$\begin{aligned}
&\widehat{z}_A = \operatorname{arctanh}(r_A) \\
&\widehat{z}_A^{\mathrm{L}} = \widehat{z}_A - z_{1-\alpha/2}\,\frac{1}{\sqrt{N_{\text{eff}} - 3}}, \quad \widehat{z}_A^{\mathrm{U}} = \widehat{z}_A + z_{1-\alpha/2}\,\frac{1}{\sqrt{N_{\text{eff}} - 3}}.
\end{aligned}$$

Transform back to the correlation scale using

$$\widehat{\rho}_A^{\mathrm{L}} = \tanh\big(\widehat{z}_A^{\text{low}}\big), \qquad \widehat{\rho}_A^{\mathrm{U}} = \tanh\big(\widehat{z}_A^{\text{high}}\big),$$

to give the familiar Fisher–z confidence interval for ρ_A, $[\widehat{\rho}_A^{\mathrm{L}}, \widehat{\rho}_A^{\mathrm{U}}]$.

[11] N_{eff} is the realized number of observations in event A, and thus corresponds to treating the conditional sample as if it were drawn directly from a bivariate normal distribution with correlation ρ_A.

8.2 Conditional Variance Only Intervals

I define a $(1-\alpha)$ CI for the LE estimator, treating unconditional correlation and variance as known, as

$$\widehat{\rho}_A^{LE} \pm z_{1-\alpha/2}\,\mathrm{SD}\big(\widehat{\rho}_A^{LE}\big),$$

where

$$\mathrm{SD}(\widehat{\rho}_A^{LE}) \approx \sqrt{\frac{\kappa_A - \frac{N_{\text{eff}}-3}{N_{\text{eff}}-1}}{N_{\text{eff}}}}\;\frac{|\rho|(1-\rho^2)}{2\sigma_A^2}\left(\rho^2 + \frac{1-\rho^2}{\sigma_A^2}\right)^{-3/2}.$$

Compare this interval with the Fisher z-interval, which reflects all sampling noise and has

$$\mathrm{SD}_{\text{Fisher}}(\widehat{\rho}_A) \approx \frac{1-\rho^2}{\sqrt{N_{\text{eff}}}}.$$

In practice, the LE interval is typically tighter than the Fisher interval. Rewriting the standard deviation of the LE interval in terms of that of the Fisher interval and a multiplicative factor gives

$$\mathrm{SD}(\widehat{\rho}_A^{LE}) = \mathrm{SD}_{\text{Fisher}}(\widehat{\rho}_A) \times \left[\frac{|\rho|}{2\sigma_A^2}\left(\rho^2 + \frac{1-\rho^2}{\sigma_A^2}\right)^{-3/2}\sqrt{\kappa_A - \frac{N_{\text{eff}}-3}{N_{\text{eff}}-1}}\right].$$

When the term in brackets is < 1, it shrinks the standard deviation and tightens the LE interval. Moreover, the LE estimator takes unconditional ρ as known, thus eliminating sampling noise from estimating $\hat{\rho}$.

I perform Monte Carlo simulation to assess Fisher intervals under conditioning and note that Fisher–z intervals track the conditional sampling distribution of r_A remarkably well across a wide range of (ρ, c) and event types. For interior events where $P(A)$ and $\overline{N}_{\text{eff}}$ are large, coverage is close to the nominal 95% level and the empirical quantiles of r_A align closely with the theoretical Fisher quantiles. For tail events, the coverage tends to be slightly conservative (95–99%), especially for large $|\rho_A|$ and moderate effective sample sizes (e.g. $c = 2$). This reflects skewness and kurtosis in the conditional distribution of r_A that Fisher's approximation only partially corrects. In our most extreme tail case ($c = 3$ with $\overline{N}_{\text{eff}} \approx 27$), the Fisher intervals retain near-nominal or slightly above-nominal coverage, illustrating their robustness as a practical tool for conditional correlation inference[12].

9 Applications:

I turn to applications of the Loretan–English (LE) conditional correlation framework. First, I study an ostensibly similar but fundamentally different phenomenon, the emergence of spurious linear relationship in ordinary least squares when regressing independent variables after conditioning on a joint event. This mechanism is *not* the LE effect (the theorem applies to conditioning on events in a single variable), but it illustrates how conditioning can severely distort correlation and

[12] See Appendix E for details of the procedure, results, and analysis of over coverage.

regression diagnostics. Second, I apply the LE theorem itself to historical financial data, using equity, bond, and FX returns, compare the LE-implied conditional correlations with empirical intra-quarter correlations, and examine divergence.

9.1 OLS Under Joint Event Conditioning

Let $\{(X_i, Y_i)\}_{i=1}^n$ be i.i.d. draws from two independent standardized Student-t distributions with three degrees of freedom:

$$X_i \sim t_3, \qquad Y_i \sim t_3, \qquad \mathrm{Cov}(X_i, Y_i) = 0, \qquad \mathrm{Var}(X_i) = \mathrm{Var}(Y_i) = 1.$$

Thus the population correlation is $\rho = 0$, and the population regression of Y on X,

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

has slope $\beta_1 = 0$ and $R^2 = 0$. Consider the joint event

$$A_t = \{X_i - Y_i > t\}, \quad t \in \{0.0, 0.5, 1.0, 1.5, 2.0\},$$

which selects samples below the line $Y = X - t$ in the (X, Y) plane. Although (X_i, Y_i) is unconditionally symmetric and independent, the selected region is increasingly asymmetric with larger t and depends jointly on both variables. I hypothesize that this very asymmetry induces nonzero conditional covariance and correlation:

$$\mathrm{Cov}(X_i, Y_i \mid X_i - Y_i > t) \neq 0, \qquad \rho_{A_t} = \mathrm{Corr}(X_i, Y_i \mid X_i - Y_i > t) > 0.$$

and I test the hypothesis using OLS and simulation. Because X_i and Y_i are standardized, the population OLS slope and R^2 in the selected sample conveniently satisfy

$$\beta_{1,A_t} = \frac{\mathrm{Cov}(X_i, Y_i \mid A_t)}{\mathrm{Var}(X_i \mid A_t)} = \rho_{A_t} \frac{\sigma_{Y|A_t}}{\sigma_{X|A_t}} \propto \rho_{A_t}, \qquad R^2_{A_t} = \rho^2_{A_t}.$$

thus I use OLS output to test whether selection on $X - Y > t$ introduces a genuine (beyond sampling error) but purely conditional linear relationship despite unconditional independence. The simulation procedure is as follows.

1. Generate $n = 200{,}000$ i.i.d. draws from the joint distribution of (X_i, Y_i).
2. For each threshold t, form the event A_t and randomly select a fixed subsample of $N_{\text{eff}} = 10{,}000$ survivors to keep regression diagnostics comparable. Note that the effective selection probability $\mathbb{P}(A_t)$ decreases as t increases.
3. For each A_t, compute the sample correlation $\hat{\rho}_{A_t}$, the estimated slope $\hat{\beta}_{1,A_t}$, its t-statistic and p-value, and the corresponding $R^2_{A_t}$.

Event	n	$\hat{\rho}_A$	$\hat{\beta}_{1,A}$	$t_{\hat{\beta}}$	$p_{\hat{\beta}}$	R^2_A
Full sample	10000	-0.00077	-0.00082	-0.07671	0.93886	0.00000
$D = X - Y > 0.0$	10000	0.31126	0.29564	32.74917	0.00000	0.09688
$D = X - Y > 0.5$	10000	0.37591	0.37133	40.56181	0.00000	0.14131
$D = X - Y > 1.0$	10000	0.40813	0.36755	44.70118	0.00000	0.16657
$D = X - Y > 1.5$	10000	0.46124	0.47025	51.97886	0.00000	0.21274
$D = X - Y > 2.0$	10000	0.52710	0.52300	62.02043	0.00000	0.27784

Table 1: OLS diagnostics for Y on X, selecting on $D_i = X_i - Y_i > t$. Unconditionally, $\rho = \beta_1 = 0$.

The full-sample regression shows essentially zero correlation, slope, and R^2, as expected. Under selection, however, the estimated correlation and slope increase almost monotonically with t, and R^2 approaches 30% for $t = 2.0$. Merely conditioning on $X > Y$ produces a statistically significant coefficient of 0.295. The LE theorem predicts a conditional correlation of 0, regardless of the univariate conditioning event, as the true ρ is 0.

However, I argue the conditional correlation framework can be extended for simple joint events such as $X - Y > t$. Define

$$D = X - Y, \qquad Z = X.$$

Because X and Y are independent with unit variance,

$$\mathrm{Var}(Z) = \mathrm{Var}(X) = 1, \qquad \mathrm{Var}(D) = \mathrm{Var}(X - Y) = 2, \qquad \mathrm{Cov}(D, Z) = \mathrm{Cov}(X - Y, X) = 1,$$

and hence the unconditional correlation between D and Z is

$$\rho_{DZ} = \mathrm{Corr}(D, Z) = \frac{\mathrm{Cov}(D, Z)}{\sqrt{\mathrm{Var}(D)}\sqrt{\mathrm{Var}(Z)}} = \frac{1}{\sqrt{2}}.$$

The OLS selection event is $A_t = \{D > t\}$ with conditional variance $\mathrm{Var}(D \mid A_t)$. Thus

$$\begin{aligned}\rho_{A_t} &= \rho_{DZ}\left[\rho_{DZ}^2 + (1 - \rho_{DZ}^2)\frac{\mathrm{Var}(D)}{\mathrm{Var}(D|A_t)}\right]^{-1/2} \\ &= \frac{1}{\sqrt{2}}\left[\frac{1}{2} + \cdot\frac{1}{\mathrm{Var}(D \mid D > t)}\right]^{-1/2}\end{aligned}$$

As stated before, for any valid right-tail event, $A_t = \{D > t\}$, the conditional correlation is defined and when $\mathrm{Var}(D \mid D > t) > \mathrm{Var}(D)$, conditional correlation exceeds unconditional correlation. Thus I confirm our hypothesis and obtain a formal explanation for the non-zero regression slopes and R^2 observed in the Monte Carlo experiment.

9.2 Empirical Intra-Quarter Conditional Correlations

Loretan and English's central contribution is to apply the conditional correlation framework to historical daily returns for pairs of securities, and argue that observed empirical correlation is consistent with changing conditional variance, and thus not necessarily reflective of changing unconditional correlation. I extend their approach to recent data, and compare model-implied conditional correlation ρ_a^{LE} with realized within-quarter correlations $\hat{\rho}_a$ to assess how much of the observed time variation in correlation can plausibly be attributed to volatility conditioning alone.

I use daily data, spanning 1985 - 2025, of four asset pairs[13]:

1. U.S. Equities vs. Rates: S&P 500 returns vs. change in the 10-year U.S. Treasury yield.[14]
2. U.K. Equities vs. Rates: FTSE 100 price index returns vs. change 10-year U.K. yield
3. German Equities vs. Rates: DAX index returns vs. change in 10-year German yield
4. FX-Equity cross: USD/JPY daily returns vs. DAX index returns

and compute one day log or percentage returns. For bond yields, I use either the first difference of the quoted yield (U.S.) or a sign-flipped bond ETF return as a yield change proxy (U.K. and Germany). After merging each pair into a common daily sample and dropping missing observations, I assign each observation to a calendar year and quarter (year, quarter). Per quarter, $N_{\text{eff}} \approx 60$.

Table 2: Sample Ranges

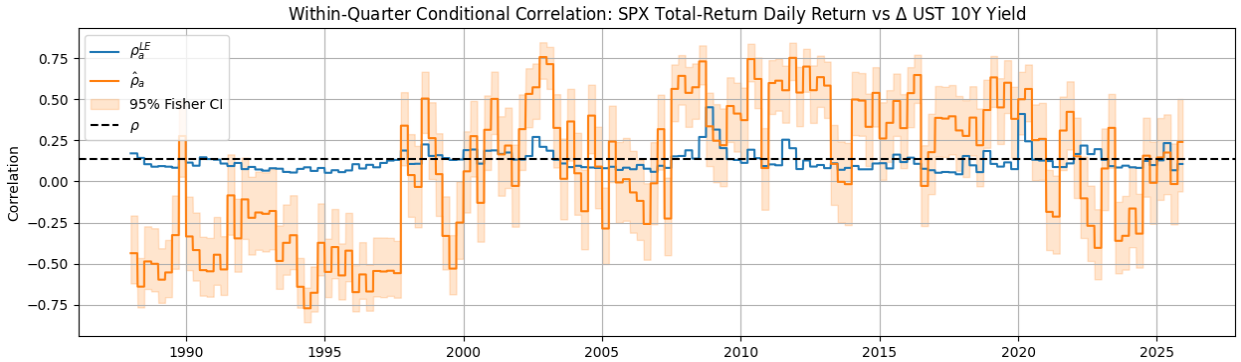
Pair	Start Date	End Date	T
SPX Daily Return vs. ΔUST-10Y Yield	1988–01–05	2025–12–04	9,473
FTSE Daily Return vs. ΔUK-10Y Yield (ETF proxy)	2008–01–03	2025–12–05	4,528
DAX Daily Return vs. ΔGerman-10Y Yield (ETF proxy)	2012–05–09	2025–12–05	3,441
USD/JPY Daily Return vs. DAX Daily Return	1996–11–01	2025–12–05	7,339

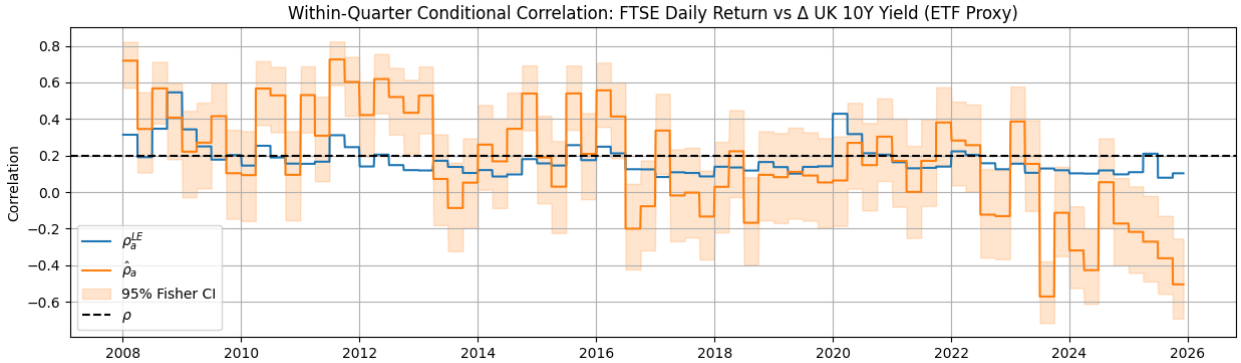
For each pair I compute the unconditional sample correlation ρ and variance $\text{Var}(X)$ over the full sample, the within-quarter variance $\text{Var}(X \mid A)$ of the equity (or FX) leg X, the within-quarter sample correlation $\hat{\rho}_a = \text{Corr}(X, Y \mid A)$, where A indexes a specific yearquarter, the Loretan-English predicted conditional correlation ρ_A^{LE}, obtained by plugging ρ, $\text{Var}(X)$ and $\text{Var}(X \mid A)$ into

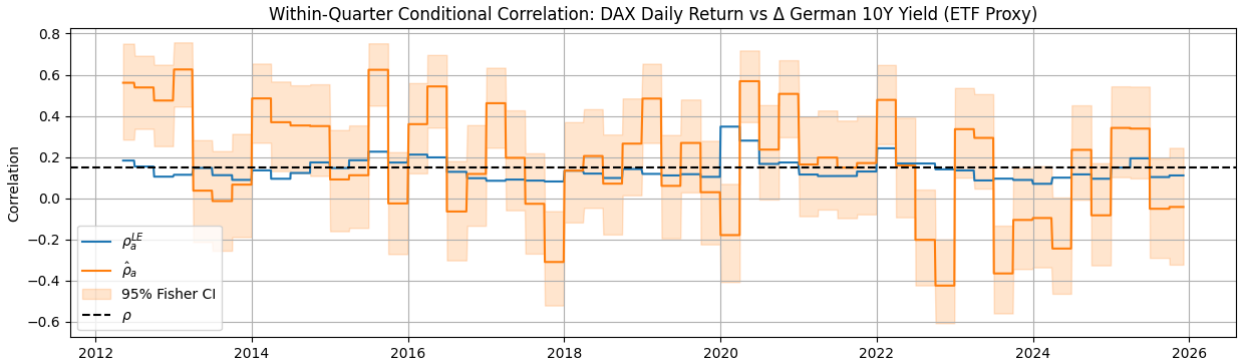
[13] The choice of asset pairs follows Loretan and English (2000) and adds the UST-10Y vs. SP500 pair, which is canonical in studies of cross-asset correlation. Equity-rate pairs capture the 'risk-on/risk-off' interaction between equity risk premia and interest rates, and each pair (U.S., U.K., German) corresponds to a large, liquid market with deep historical coverage and well-documented structural episodes such as monetary regime shifts, inflation cycles, and crisis periods. Regarding bonds, the U.S. U.K. and German 10-year benchmarks represent the most liquid sovereign curves globally, making them natural proxies for discount-rate shocks. For FX, the USD/JPY–DAX pair represents an FX–equity cross driven by global risk sentiment, capital flows, and carry-trade dynamics. USD/JPY a highly liquid currency pair with well known 'safe-haven' behavior. These four pairs provide broad coverage across asset classes, regions, and economic mechanisms, while maintaining long, high-frequency time series suitable for analysis.

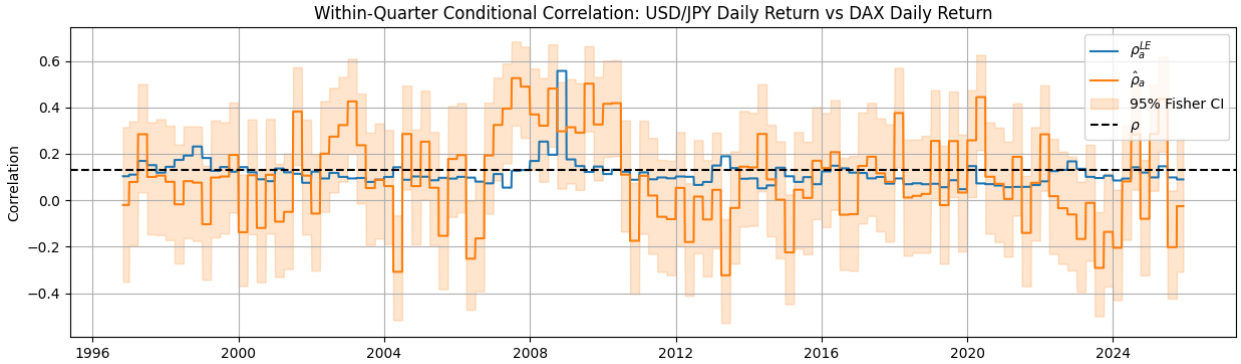
[14] I obtain all data from Yahoo Finance, under the following tickers: `^SP500TR` (S&P 500 Total Return index), `^FTSE` (FTSE 100), `^GDAXI` (DAX), `^JPY=X` (USD/JPY spot rate), `IGILT.L` (UK 10Y Gilt ETF), `IS0L.DE` (German 10Y Bund ETF). The UST-10Y yield is `DGS10`, obtained from FRED.

the LE formula, and a 95% Fisher-z confidence interval. For each pair, I plot ρ_A^{LE} (blue), empirical $\hat{\rho}_A$ (orange), the 95% Fisher interval (orange), and the unconditional correlation ρ (black).









Note the following. First, across all four pairs, the LE conditional correlation ρ_a^{LE} is remarkably smooth and remains close to the unconditional correlation ρ. Naturally, the curve evolves slowly, as conditional correlation is a deterministic function of the ratio of unconditional to conditional variance. In each market, the equity or FX leg does exhibit time-varying quarterly volatility, but the fluctuations are not large or persistent enough to generate the dramatic correlation swings observed in the realized within-quarter correlations, $\hat{\rho}_a$ (orange).

Second, the realized within-quarter correlations $\hat{\rho}_a$ display large, sometimes abrupt regime-like movements. They flip signs and push towards ± 0.8. These movements are especially pronounced for the SPX-UST and USD/JPY-DAX pairs, and correspond to flight-to-quality episodes, risk-on/risk-off dynamics, and major macro shocks (e.g. LTCM 1998, Dot-Com 2000, GFC 2007-2009, Eurozone 2011, Covid 2020).

Third, the 95% Fisher confidence bands are fairly wide (on the order of ± 0.3), indicating that a single quarter provides a noisy estimate of the underlying correlation. However, for many quarters the LE curve and the unconditional correlation fall *outside* the 95% intervals implied by the observed $\hat{\rho}_a$. This pattern is most prominent during crisis or transition periods, when the orange curve jumps to values that are statistically incompatible with a constant correlation plus volatility driven adjustment.

Fourth, and critically, the LE theorem cannot explain the changes in the sign of within-quarter correlation. Per 1, ρ_a is necessarily the same sign as ρ. Conditional variance inflates or deflates *magnitude*, not *direction*. The drastic change for the UST-10Y vs. SPX pair (largely negative over 1990-2000, largely positive from 2000 onward) indicates the theorem doesn't hold.

To examine why the LE theorem fails, I evaluate the data's time series properties[15]. I find that across all four empirical asset pairs, some of the theorem's implicit assumptions (homoskedastic errors, covariance stationarity, absence of serial correlation) hold to varying degrees and some completely fail. While all series are stationary, the data exhibits strong heteroskedasticity, pronounced volatility clustering, heavy tails and non-elliptical distributions, and in some cases residual autocorrelation. These results indicate that while the LE mechanism provides a useful volatility standardized benchmark, the bulk of time variation in within-quarter financial correlations reflects genuine changes in the underlying joint distribution, rather than an artifact of changing volatility or finite-sample noise. Conditional volatility is not the sole driver of correlation variation. Instead, genuine time-varying economic relationships, structural shocks, risk-on/risk-off regimes, and tail asymmetries play a decisive role in explaining the large, regime-dependent swings observed in empirical within-quarter conditional correlations.

10 Conclusion

I extend and evaluate the conditional-correlation framework of Boyer et al. (1997) and Loretan and English (2000), providing theoretical, computational, and empirical insights into when volatility

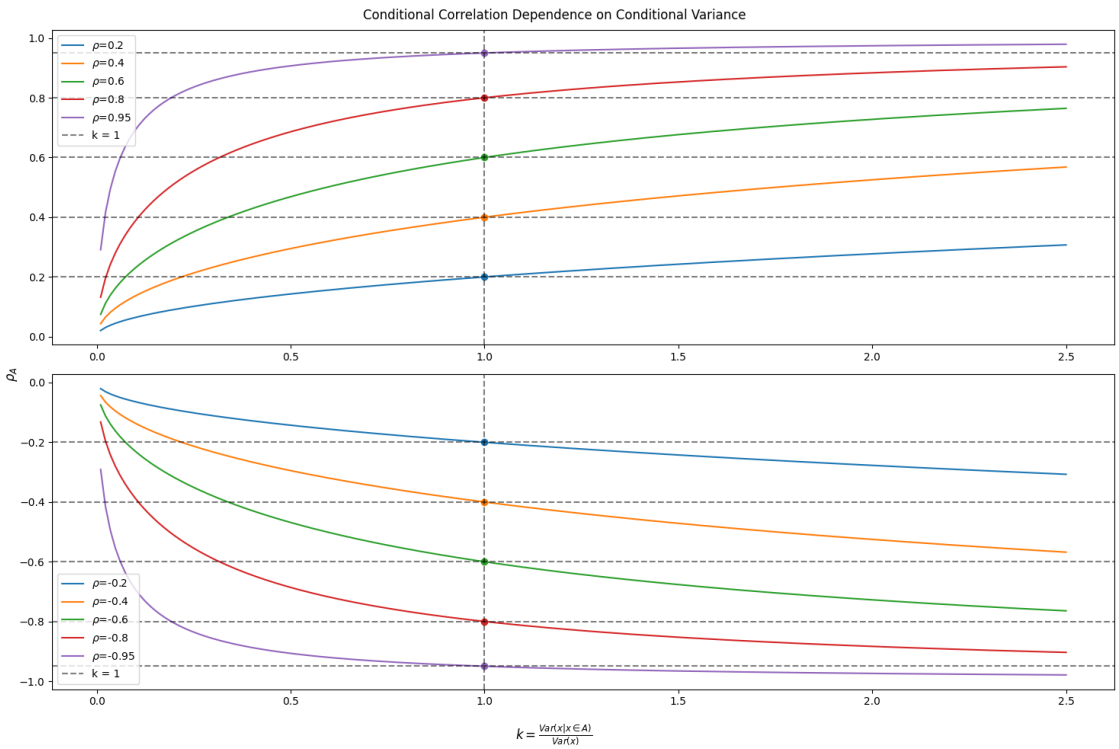
[15] See Appendix F for the complete test results and analysis.

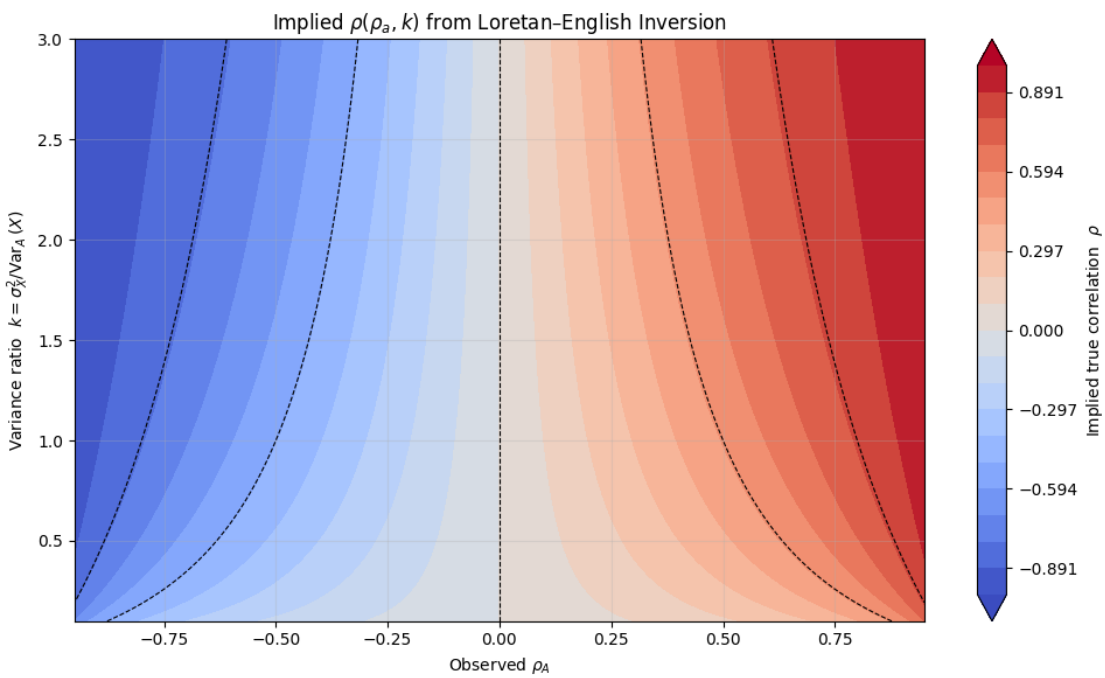
driven distortions meaningfully affect observed correlations. I demonstrate the following.

First, the LE expression for conditional correlation can be derived using a regression based framework, without relying on bivariate normality. The theorem has intuitive implications. Conditioning on high-variance subsamples inflates correlations toward ± 1, while conditioning on low-variance subsamples pushes them to zero. Second, I provide an inversion formula to compute unconditional correlation from observed conditional correlation and conditional variance. Using simulation, I identify regions where inference is fragile, particularly near $\rho = 0$. Third, I derive the sampling variance of the conditional variance estimator and perform Monte Carlo experiments which show the LE mechanism is numerically stable for an extreme tail event over most of the correlation range. Fourth, I show conditional correlation estimates are unavoidably noisy for the small subsamples typical of tail events. Fifth, I derive Fisher-z confidence intervals, demonstrate their robustness even under strong conditioning, and show that LE-based intervals which assume only variability from conditional variance are tighter. Finally, I apply the conditional correlation framework to OLS, demonstrating that it can explain distortions that arise from conditioning on joint events, and to an updated set of asset return time series, demonstrating that across equity-rate and FX-equity pairs in the U.S., U.K., and Germany, realized within-quarter correlations display large, abrupt shifts that cannot be explained by volatility conditioning alone.

Our results enhance and refine the theoretical framework for conditional correlation, and demonstrate that while volatility conditioning is a powerful explanatory mechanism, the correlation dynamics of real world assets overwhelmingly reflect true structural variation, meaning a change in the underlying joint distribution, rather than the statistical artifact of conditional variance.

11 Figures


Conditional Correlation Dependence on Conditional Variance
ρ=0.2
ρ=0.4
ρ=0.6
ρ=0.8
ρ=0.95
k = 1
ρ=-0.2
ρ=-0.4
ρ=-0.6
ρ=-0.8
ρ=-0.95
k = 1
ρA
k = Var(x|x∈A) / Var(x)


Implied ρ(ρa, k) from Loretan–English Inversion
Variance ratio k = σ²X/VarA (X)
Observed ρA
Implied true correlation ρ
0.891
0.594
0.297
0.000
−0.297
−0.594
−0.891

Appendix

A Monte Carlo LE and Inversion Validation

I simulate $(X, Y) \sim N_2(0, \Sigma)$ with $\mathrm{Var}(X) = \mathrm{Var}(Y) = 1$ and $\mathrm{Corr}(X, Y) \in \{0.5, 0.95\}$. I form conditional subsamples using decile-delimited intervals of X and absolute-tail events $A = \{|X| \geq q\}$, where both decile cutpoints and tail thresholds are empirical quantiles of the simulated X sample. I use $n = 10,000$ unconditonal draws, and for each event I report the subsample size N_{eff}, the conditional variance $\mathrm{Var}(X \mid A)$ and variance ratio k, the empirical conditional correlation $\hat{\rho}_A$, the Loretan-English prediction ρ_A^{LE}, and the inverted correlation estimate ρ_{inv}. Small discrepancies between empirical and analytic quantities reflect finite-sample Monte Carlo noise, particularly in the most extreme tails where N_{eff} is small and the conditional variance is estimated imprecisely.

A.1 Decile-Delimited Events

Decile	Interval	N_{eff}	k	$\mathrm{Var}(x \in D_i)$	$\hat{\rho}_A$	ρ_A^{LE}	ρ_{inv}
1	$(-3.703, -1.307)$	1000	6.21335	0.16100	0.19465	0.22565	0.44338
2	$(-1.307, -0.856)$	1000	61.57183	0.01600	0.05738	0.07338	0.41111
3	$(-0.856, -0.537)$	1000	115.78259	0.00900	0.05451	0.05358	0.50649
4	$(-0.537, -0.272)$	1000	171.28923	0.00600	0.04833	0.04407	0.53502
5	$(-0.272, -0.022)$	1000	208.19217	0.00500	0.02348	0.03998	0.32096
6	$(-0.022, 0.234)$	1000	179.82924	0.00600	0.03382	0.04301	0.41322
7	$(0.234, 0.492)$	1000	185.05079	0.00500	0.07134	0.04240	0.69734
8	$(0.492, 0.815)$	1000	117.35879	0.00900	0.04725	0.05322	0.45602
9	$(0.815, 1.280)$	1000	55.97425	0.01800	0.05937	0.07694	0.40656
10	$(1.280, 3.458)$	1000	6.43181	0.15600	0.23946	0.22197	0.53031

Table 3: Decile subsample results for $\rho = 0.5$ (full-sample correlation $\hat{\rho} \approx 0.501$).

Decile	Interval	N_{eff}	k	$\text{Var}(x \in D_i)$	$\hat{\rho}_A$	ρ_A^{LE}	ρ_{inv}
1	$(-3.856, -1.306)$	1000	5.94005	0.17000	0.78648	0.78046	0.95181
2	$(-1.306, -0.861)$	1000	61.90644	0.01600	0.33074	0.36066	0.94009
3	$(-0.861, -0.529)$	1000	119.97738	0.00800	0.22911	0.26763	0.93232
4	$(-0.529, -0.243)$	1000	150.47567	0.00700	0.25413	0.24073	0.95509
5	$(-0.243, 0.007)$	1000	195.76822	0.00500	0.18261	0.21248	0.93329
6	$(0.007, 0.260)$	1000	183.43769	0.00600	0.22053	0.21917	0.95060
7	$(0.260, 0.525)$	1000	173.62977	0.00600	0.21785	0.22497	0.94677
8	$(0.525, 0.836)$	1000	125.90569	0.00800	0.29335	0.26169	0.96032
9	$(0.836, 1.279)$	1000	65.35613	0.01500	0.36172	0.35222	0.95275
10	$(1.279, 3.276)$	1000	6.13523	0.16500	0.76523	0.77549	0.94688

Table 4: Decile subsample results for $\rho = 0.95$ (full-sample correlation $\hat{\rho} \approx 0.949$).

A.2 Tail Events

q	Threshold	N_{eff}	k	$\text{Var}(x \in A)$	$\hat{\rho}_A$	ρ_A^{LE}	ρ_{inv}
0.80000	1.29286	2000	0.30592	3.27100	0.71547	0.72211	0.49285
0.90000	1.66233	1000	0.22855	4.37800	0.76713	0.77022	0.49631
0.95000	1.97995	500	0.18255	5.48100	0.79730	0.80383	0.49151
0.97500	2.23505	250	0.15049	6.64800	0.83944	0.83003	0.51401
0.99000	2.56215	100	0.12234	8.17900	0.84525	0.85529	0.48409

Table 5: Tail subsample results for $\rho = 0.5$ (full-sample correlation $\hat{\rho} \approx 0.501$). Threshold is the empirical q-quantile of $|x|$.

q	Threshold	N_{eff}	k	$\text{Var}(x \in A)$	$\hat{\rho}_A$	ρ_A^{LE}	ρ_{inv}
0.80000	1.29305	2000	0.30833	3.28300	0.98342	0.98375	0.94903
0.90000	1.64761	1000	0.22792	4.44100	0.98745	0.98791	0.94822
0.95000	1.97999	500	0.17996	5.62500	0.99020	0.99042	0.94892
0.97500	2.26936	250	0.14861	6.81100	0.99180	0.99207	0.94842
0.99000	2.57840	100	0.12138	8.33900	0.99397	0.99351	0.95336

Table 6: Tail subsample results for $\rho = 0.95$ (full-sample correlation $\hat{\rho} \approx 0.949$). Threshold is the empirical q-quantile of $|x|$.

B Computational Cost of Rejection Sampling

Robert and Casella [9, Example 2.20] argue rejection sampling is naive on account of the computational cost. The procedure is to draw $Z \sim p$ from the unconditional distribution and accept it

only if $Z \in A$, thus each proposal is accepted with probability $p_A = \Pr(Z \in A)$. Thus the number of proposals required to obtain the first accepted draw is geometric with mean $1/p_A$, and obtaining M accepted samples has expected cost

$$\mathbb{E}[N_{\text{proposals}}] = \frac{M}{p_A}.$$

For rare events, the approach is cost-prohibitive. If $X \sim N(0,1)$ and $A = \{X > 3\}$, then $p_A \approx 0.00135$, and obtaining 10,000 accepted draws requires about 7.4×10^6 proposals on average. Programmatically, rejectiong sampling requires while loops, which is inadvisable for many independent Monte Carlo replications even with stopping conditions. See the algorithm below:

1. Initialize an empty collection of draws $X \leftarrow \emptyset$. Set the accepted sample counter $k \leftarrow 0$ and the proposal counter $t \leftarrow 0$.
2. While $k < M$ and $t <$ max_iter:
   (a) Draw a proposal z from the unconditional distribution p.
   (b) Increment the proposal counter: $t \leftarrow t + 1$.
   (c) If $z \in A$, then
      i. Append z to the collection X.
      ii. Increment the accepted sample counter: $k \leftarrow k + 1$.
3. Return the vector of accepted draws X.

C Conditional Variance and Higher Moments Under Truncation

I verify that our analytic conditional variance expressions are reproduced by large simulation runs and examine the sampling variability of the estimator under interior and tail truncations. Throughout this section, I assume $X \sim N(0,1)$ and consider symmetric interior events $A = \{|X| < c\}$ and tail events $A = \{|X| > c\}$ unless otherwise noted.

C.1 Numerical Evaluation of Conditional Variance

As a consistency check for our derived tail event conditional variance expression and the truncated normal distribution's closed form expressions for $\mathrm{Var}(X \mid A)$, I compare output from `scipy.stats.truncnorm`'s implementation and the Monte Carlo estimate from a single large i.i.d. simulation. Doing so checks whether a single massive sample produces an estimate that converges to the population value.

Table 3: Analytic vs. Single Run Monte Carlo Conditional Variance

Case	Event	$P(A)$	Method	Estimated $\mathrm{Var}(X \mid A)$
Tail	$\{\lvert X\rvert > 1\}$	$2(1-\Phi(1)) \approx 0.3173$	Analytic	2.525135
			Monte Carlo ($n = 10^7$)	2.525383
Interior	$(-1, 1)$	$2\Phi(1) - 1 \approx 0.6827$	Closed form	0.291125
			`scipy.stats.truncnorm`	0.291125
			Monte Carlo ($n = 10^7$)	0.291238

Use 1 unconditional draw of size $n = 10^7$ per estimate, where $N_{\text{eff}} \approx nP(A)$.

For both events, the analytic and MC values are quite close, confirming the conditional variance formulas and effectiveness of large single-run simulations.

C.2 Conditional Kurtosis and Relative Standard Deviation

Computing RSD requires computing conditional variance σ_A^2 and conditional kurtosis κ_A. Note than when the event is symmetric around zero, $\mathbb{E}[X \mid A] = 0$, thus variance and kurtosis depend only on the second and fourth moments. I outline derivation techniques.

Interior Events: For $A = \{|X| < c\}$, the truncated-normal distribution yields closed-form expressions for $\mathbb{E}[X^2 \mid A]$ and $\mathbb{E}[X^4 \mid A]$, which are available directly from `scipy.stats.truncnorm`. From these values, σ_A^2 and κ_A follow immediately.

Tail Events: For $A = \{|X| > c\}$, I obtain conditional moments using the law of total expectation. In one form, it states for finite or countable paritions of the sample space,

$$\mathbb{E}(Y) = \sum_i \mathbb{E}(Y|A_i)P(A_i)$$

Letting $Y = X^r$ and noting events $\{|X| > c\}$ and $\{|X| < c\}$ partition the support of X,

$$\mathbb{E}[X^2] = P(|X| < c)\,\mathbb{E}[X^2 \mid |X| < c] + P(|X| > c)\,\mathbb{E}[X^2 \mid |X| > c],$$

$$\mathbb{E}[X^4] = P(|X| < c)\,\mathbb{E}[X^4 \mid |X| < c] + P(|X| > c)\,\mathbb{E}[X^4 \mid |X| > c].$$

Since $\mathbb{E}[X^2] = 1$ and $\mathbb{E}[X^4] = 3$ for the standard normal, the tail moments follow by solving for $\mathbb{E}[X^2 \mid |X| > c]$ and $\mathbb{E}[X^4 \mid |X| > c]$. These yield σ_A^2 and κ_A for both interior and tail cases.

C.3 Monte Carlo Evaluation of Variance Estimator Accuracy

I now examine the finite-sample accuracy of the conditional variance estimator. For each cutoff c and unconditional sample size n, I run R independent Monte Carlo replications:

1. Draw n samples from $N(0, 1)$

2. Filter according to the event A

3. Compute the sample variance of the retained points, producing one estimate $S^2_{A,r}$.

Across replications, I record the MC mean, standard error, and effective sample size. Note that $\mathrm{RSD}(S^2_A) \propto 1/\sqrt{N_{\text{eff}}}$, thus larger $P(A)$ yields larger N_{eff} and more stable estimates. In the tables below, each row shows the analytic conditional variance, the MC mean across $R = 100$ replications, the MC standard error, the average effective sample size, and the closed-form RSD.

Table 7: Monte Carlo Estimate of Conditional Variance, Interior Events (Multiple Runs)

c	$P(A)$	n	Analytic Var	Mean Var (MC)	SE (MC)	$\overline{N}_{\text{eff}}$	RSD (CF)
0.5	0.3829	10^2	0.080589	0.078504	0.001170	38	0.1529
0.5		10^6	0.080589	0.080589	0.000011	382941	0.0015
1.0	0.6827	10^2	0.291125	0.282612	0.003467	68	0.1195
1.0		10^6	0.291125	0.291120	0.000034	682667	0.0012
2.0	0.9545	10^2	0.773741	0.758811	0.009312	96	0.1202
2.0		10^6	0.773741	0.773704	0.000090	954501	0.0012

Table 8: Monte Carlo Estimate of Conditional Variance, Tail Events (Multiple Runs)

c	$P(A)$	n	Analytic Var	Mean Var (MC)	SE (MC)	$\overline{N}_{\text{eff}}$	RSD (CF)
0.5	0.6171	10^2	1.570539	1.554585	0.019598	63	0.1260
0.5		10^6	1.570539	1.570578	0.000195	616994	0.0013
1.0	0.3173	10^2	2.525135	2.424376	0.028639	32	0.1240
1.0		10^6	2.525135	2.525180	0.000294	317276	0.0012
2.0	0.0455	10^2	5.746431	4.752238	0.187774	5	0.3455
2.0		10^6	5.746431	5.744408	0.000724	45507	0.0015

Note the following. First, since $P(|X| < c)$ is large, interior events retain many samples even at moderate n. The conditional variance estimator stabilizes quickly, and MC estimates match analytic values to several decimal places once $n \geq 10^4$, exactly as predicted by the small RSD values. Second, for tail events, $P(|X| > c)$ shrinks rapidly, which reduces N_{eff} and increases estimator variability. Moderate tails such as $c = 1$ remain tractable, but more extreme tails such as $c = 2$ require very large n (up to 10^6) for the MC mean and standard error to align with analytic values at even two decimal places. Large RSD values reflect small survival probabilities and the conditional distribution's elevated kurtosis.

D Conditional Correlation Sampling Variance

D.1 Monte Carlo Evaluation of LE Sampling Variance

To evaluate the delta-method approximation of the conditional correlation estimator's finite-sample variability, I perform Monte Carlo simulation using $(X, Y) \sim N_2(0, 1)$ with unconditional correlation $\rho = 0.5$ and the tail event $A = \{|X| > 2\}$ which has $P(A) \approx 0.0455$. The corresponding conditional variance and kurtosis of X are $\sigma_A^2 \approx 5.7464$ and $\kappa_A \approx 1.10$, respectively. I target $N_{\text{eff}} \approx 60$, implying an unconditional draw size $n \approx N_{\text{eff}}/P(A) \approx 1319$.

For each of $R = 1000$ MC replications, I draw n observations from the bivariate normal, filter on the event A, compute the conditional sample variance S_A^2 of X, and form $\widehat{\rho}_A^{LE}$. I also record the realized $N_{\text{eff},r}$ to confirm alignment with the target survival count. Using our delta-method derived sampling standard deviation and the corresponding population LE correlation, I find

$$\begin{aligned}
\text{SD}(\widehat{\rho}_A^{LE}) &\approx \sqrt{\frac{\kappa_A - \frac{N_{\text{eff}}-3}{N_{\text{eff}}-1}}{N_{\text{eff}}}} \cdot \frac{|\rho|(1-\rho^2)}{2\sigma_A^2}\left(\rho^2 + \frac{1-\rho^2}{\sigma_A^2}\right)^{-3/2} \\
&\approx \sqrt{\frac{1.10 - \frac{60-3}{60-1}}{60}} \cdot \frac{|0.5|(1-0.25)}{2 \cdot 5.7464}\left(0.25 + \frac{1-0.25}{5.7464}\right)^{-3/2} = 0.00649 \\
\rho_A^{LE} &= \frac{\rho}{\sqrt{\rho^2 + (1-\rho^2)/\sigma_A^2}} = \frac{0.5}{\sqrt{0.25 + (1-0.25)/5.7464}} \approx 0.8107.
\end{aligned}$$

The table below compares analytic predictions with those from the Monte Carlo procedure.

Table 9: Monte Carlo Estimate of LE Conditional Correlation Sampling Standard Deviation

c	ρ	$P(A)$	n	N_{eff}^*	ρ_A^{LE} (analytic)	Mean $\widehat{\rho}_A^{LE}$ (MC)	SD($\widehat{\rho}_A^{LE}$)
2.0	0.5	0.0455	1319	60	0.8107	0.8108	MC: 0.00646 Analytic: 0.00649

MC results confirm the accuracy of the delta-method approximation. The empirical standard deviation (0.00646) matches the analytic value (0.00649) to within 5×10^{-5}, and the MC mean aligns closely with the population conditional correlation. Even in a low-survival tail event, our closed form variance describes the LE sampling distribution well.

D.2 Effect of Tail Severity on LE Sampling Variability

I provide the complete details for the cited example demonstrating how the standard deviation of $\widehat{\rho}_A^{LE}$ changes as the truncation becomes more extreme, holding the effective sample size fixed. Let (X, Y) be bivariate normal with correlation $\rho = 0.5$, and consider the symmetric tail events

$$A_c = \{|X| > c\}, \qquad c \in \{2, 3\}.$$

Fix $N_{\text{eff}} = 60$. I go through each component of $\mathrm{SD}(\widehat{\rho}_A^{LE})$ to draw out the conclusion. The conditional variance and conditional kurtosis values are

$$\sigma_{A_2}^2 \approx 5.746, \qquad \sigma_{A_3}^2 \approx 10.84, \qquad \kappa_{A_2} \approx 1.096, \qquad \kappa_{A_3} \approx 1.033.$$

With $(N_{\text{eff}} - 3)/(N_{\text{eff}} - 1) = 57/59 \approx 0.9661$, compute the kurtosis factors

$$\sqrt{\frac{\kappa_{A_2} - 0.9661}{60}} \approx 0.0466, \qquad \sqrt{\frac{\kappa_{A_3} - 0.9661}{60}} \approx 0.0334.$$

Next, compute the variance scaling factors

$$\frac{|\rho|(1-\rho^2)}{2\sigma_{A_2}^2} \approx 0.0326, \qquad \frac{|\rho|(1-\rho^2)}{2\sigma_{A_3}^2} \approx 0.0173,$$

and finally

$$\rho^2 + \frac{1-\rho^2}{\sigma_{A_2}^2} \approx 0.3805, \qquad (0.3805)^{-3/2} \approx 4.26,$$

$$\rho^2 + \frac{1-\rho^2}{\sigma_{A_3}^2} \approx 0.3192, \qquad (0.3192)^{-3/2} \approx 5.54.$$

Putting the components together gives the final standard deviations

$$\mathrm{SD}(\widehat{\rho}_{A_2}^{LE}) \approx 0.0466 \times 0.0326 \times 4.26 \approx 0.0065,$$

$$\mathrm{SD}(\widehat{\rho}_{A_3}^{LE}) \approx 0.0334 \times 0.0173 \times 5.54 \approx 0.0032.$$

This example shows exactly how the higher truncation changes the conditional moments and leads to noticeably different sampling variability. The higher conditional variance and lower kurtosis at more $c = 3$ both contribute to the decrease in the SD of $\widehat{\rho}_A^{LE}$.

E Monte Carlo Assessment of Fisher Intervals

I examine the performance of Fisher-z intervals on conditional samples using simulation. As before, $(X, Y) \sim N(0, 1)$, but I vary unconditional correlation ρ over $\{-0.9, -0.5, 0, 0.5, 0.9\}$. For each ρ and cutoff $c \in \{1, 2, 3\}$ I consider tail events $A = \{|X| > c\}$ and interior events $A = \{|X| < c\}$. The procedure is as follows. For each triplet (event type, ρ, c):

1. Compute event probability $P(A)$ and conditional variance $\sigma_A^2 = \mathrm{Var}(X \mid A)$ using analytic formulas.

2. Compute the true conditional correlation ρ_A via the LE formula.

3. For each of $R = 1000$ replications, draw $n = 10{,}000$ i.i.d. observations from the bivariate normal, filter on event A, and compute the empirical conditional correlation r_A together with

the effective sample size N_{eff}.

4. For each r in R, construct a Fisher-z 95% confidence interval for ρ_A using the replication-specific $N_{\text{eff}}^{(r)}$ in the standard error $1/\sqrt{N_{\text{eff}}^{(r)} - 3}$, and record whether the interval contains ρ_A (1 if interval covers, 0 otherwise).

5. Over R replications, report

   - Average effective sample size $\overline{N}_{\text{eff}}$,
   - Empirical mean, standard deviation, and 2.5% / 97.5% quantiles of r_A,
   - Proportion of replications whose Fisher-z interval contains ρ_A (empirical coverage),
   - Theoretical 2.5% / 97.5% Fisher quantiles using $N_{\text{eff}} = \lceil \overline{N}_{\text{eff}} \rceil$.

The tables below show the MC performance of Fisher-z intervals for tail and interior events. I report all (ρ, c) combinations but for brevity focus on the average effective sample size, the sampling distribution of r_A, and the empirical coverage.

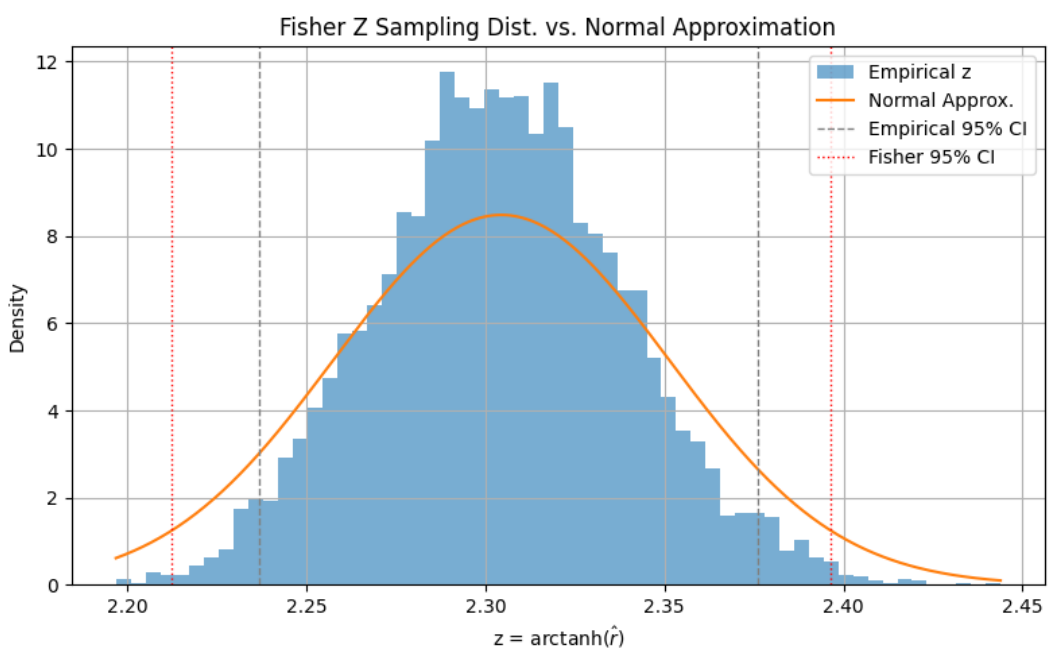
Table 10: Monte Carlo Performance of Fisher–z 95% CIs, Tail Events

ρ	c	$P(A)$	$\overline{N}_{\text{eff}}$	ρ_A	Mean	SD	Empirical r_A Quantile		Fisher Quantile		Coverage
							2.5%	97.5%	2.5%	97.5%	
	1.0	0.3173	3174.6	-0.95656	-0.95649	0.00121	-0.95881	-0.95410	-0.95942	-0.95350	0.9880
−0.90	2.0	0.0455	455.4	-0.98019	-0.98032	0.00133	-0.98274	-0.97770	-0.98350	-0.97623	0.9940
	3.0	0.0027	27.2	-0.98936	-0.98976	0.00300	-0.99466	-0.98336	-0.99519	-0.97655	0.9860
	1.0	0.3173	3173.5	-0.67604	-0.67568	0.00936	-0.69390	-0.65727	-0.69450	-0.65669	0.9560
−0.50	2.0	0.0455	455.0	-0.81056	-0.81001	0.01296	-0.83503	-0.78359	-0.83990	-0.77649	0.9850
	3.0	0.0027	27.1	-0.88509	-0.88825	0.03358	-0.94285	-0.81400	-0.94664	-0.76123	0.9830
	1.0	0.3173	3176.0	0.00000	-0.00018	0.01792	-0.03522	0.03342	-0.03478	0.03478	0.9500
0.00	2.0	0.0455	455.5	0.00000	-0.00081	0.04705	-0.09036	0.09120	-0.09188	0.09188	0.9540
	3.0	0.0027	27.1	0.00000	0.00094	0.20158	-0.38943	0.39066	-0.37932	0.37932	0.9530
	1.0	0.3173	3171.4	0.67604	0.67626	0.00876	0.65938	0.69338	0.65668	0.69450	0.9720
0.50	2.0	0.0455	455.7	0.81056	0.81102	0.01315	0.78422	0.83540	0.77651	0.83988	0.9890
	3.0	0.0027	27.1	0.88509	0.88825	0.03426	0.80479	0.94398	0.76142	0.94660	0.9790
	1.0	0.3173	3171.5	0.95656	0.95654	0.00116	0.95426	0.95884	0.95350	0.95942	0.9920
0.90	2.0	0.0455	455.9	0.98019	0.98028	0.00136	0.97753	0.98281	0.97624	0.98350	0.9960
	3.0	0.0027	26.9	0.98936	0.98980	0.00319	0.98256	0.99484	0.97645	0.99521	0.9870

Table 11: Monte Carlo Performance of Fisher–z 95% CIs, Interior Events

ρ	c	$P(A)$	$\overline{N}_{\text{eff}}$	ρ_A	Mean	SD	Empirical r_A Quantile		Fisher Quantile		Coverage
							2.5%	97.5%	2.5%	97.5%	
	1.0	0.6827	6826.1	-0.74417	-0.74406	0.00487	-0.75323	-0.73443	-0.75457	-0.73340	0.9700
−0.90	2.0	0.9545	9545.0	-0.87599	-0.87608	0.00219	-0.88067	-0.87176	-0.88058	-0.87124	0.9630
	3.0	0.9973	9973.2	-0.89767	-0.89762	0.00200	-0.90131	-0.89373	-0.90141	-0.89379	0.9520
	1.0	0.6827	6827.9	-0.29742	-0.29717	0.01146	-0.32004	-0.27510	-0.31889	-0.27564	0.9390
−0.50	2.0	0.9545	9543.9	-0.45281	-0.45274	0.00823	-0.46938	-0.43655	-0.46861	-0.43671	0.9450
	3.0	0.9973	9972.9	-0.49494	-0.49496	0.00743	-0.50901	-0.47981	-0.50962	-0.47998	0.9560
	1.0	0.6827	6826.7	0.00000	-0.00079	0.01232	-0.02597	0.02270	-0.02372	0.02372	0.9470
0.00	2.0	0.9545	9545.1	0.00000	-0.00011	0.01014	-0.01940	0.02012	-0.02006	0.02006	0.9530
	3.0	0.9973	9973.0	0.00000	0.00007	0.00989	-0.01935	0.01950	-0.01963	0.01963	0.9520
	1.0	0.6827	6827.2	0.29742	0.29711	0.01092	0.27634	0.31879	0.27564	0.31889	0.9510
0.50	2.0	0.9545	9544.8	0.45281	0.45287	0.00799	0.43726	0.46812	0.43671	0.46861	0.9530
	3.0	0.9973	9973.0	0.49494	0.49490	0.00758	0.48042	0.50924	0.47998	0.50962	0.9560
	1.0	0.6827	6824.0	0.74417	0.74408	0.00488	0.73417	0.75373	0.73340	0.75458	0.9700
0.90	2.0	0.9545	9544.6	0.87599	0.87599	0.00224	0.87148	0.88031	0.87124	0.88058	0.9580
	3.0	0.9973	9972.9	0.89767	0.89758	0.00188	0.89375	0.90145	0.89379	0.90141	0.9470

Overall, Fisher–z intervals retain near-nominal coverage under both tail and interior conditioning, even for quite small $\overline{N}_{\text{eff}}$. Slight overcoverage appears in the most extreme tail cases, where the sampling distribution of r_A (and hence z_A) is skewed. I investigate over coverage further, examining the specific case of $\rho = 0.9$ and $A = \{|X| > 2\}$ which has empirical coverage of 99.6%. I plot the sampling distribution of Fisher-z below. Note that the theoretical interval is slightly wider and



more symmetric than the empirical distribution, yielding higher coverage. The symmetric normal

approximation allocates too much mass to the lighter tail and hence produces more conservative Fisher intervals.

F Time Series Diagnostics

I apply tests for stationarity, homoskedasticity, Gaussian residuals, and serial correlation in residuals to the equity, FX, and bond return series to understand why empirical within-quarter conditional correlations diverge from the Loretan–English (LE) benchmark. Residual tests use the regression $y_t = \beta x_t + e_t$.

Table 12: ADF Tests for Stationarity

Pair	ADF(x) Stat.	ADF(x) p	ADF(y) Stat.	ADF(y) p
SPX vs. ΔUST-10Y	-16.30	$< 10^{-4}$	-21.17	$< 10^{-4}$
FTSE vs. ΔUK-10Y	-13.98	$< 10^{-4}$	-21.76	$< 10^{-4}$
DAX vs. ΔDE-10Y	-20.87	$< 10^{-4}$	-11.53	$< 10^{-4}$
USD/JPY vs. DAX	-27.06	$< 10^{-4}$	-36.51	$< 10^{-4}$

ADF tests reject the unit root for all six variables with $p \approx 0$, confirming that the underlying return series are stationary.

Table 13: Breusch–Pagan and ARCH LM Tests for Heteroskedasticity

Pair	BP Stat.	BP p	ARCH(5) Stat.	ARCH(5) p
SPX vs. ΔUST-10Y	8.11	0.0044	226.16	$< 10^{-4}$
FTSE vs. ΔUK-10Y	7.29	0.0069	636.96	$< 10^{-4}$
DAX vs. ΔDE-10Y	41.30	$< 10^{-4}$	590.03	$< 10^{-4}$
USD/JPY vs. DAX	2.04	0.1530	1045.88	$< 10^{-4}$

Breusch–Pagan tests reject homoskedasticity for three of the four pairs (fail to reject H_0 at a 5% level for USD/JPY vs. DAX), and ARCH LM test statistics are significant for all series with p–values of effectively zero. I conclude volatility is state-dependent, persistent, and clustered, matching the classic ARCH/GARCH pattern and violating the LE theorem's assumption of i.i.d errors with constant variance. Moreover, $\mathrm{Var}(u_t)$ co-moves strongly with $|x_t|$, contradicting the LE supposition that conditioning affects correlation only through $\mathrm{Var}(X \mid A)$.

Table 14: Ljung–Box Tests for Serial Correlation (h=5,10)

Pair	LB(5) p	LB(10) p
SPX vs. ΔUST-10Y	0.00010	0.00011
FTSE vs. ΔUK-10Y	0.05591	0.00013
DAX vs. ΔDE-10Y	0.15833	0.16117
USD/JPY vs. DAX	0.12030	0.33339

Ljung–Box tests show significant serial correlation for SPX-UST and at 10 lags for FTSE-UK, although DAX-DE and USDJPY–DAX show little evidence of serial correlation.

Table 15: Anderson–Darling Test for Normality

Pair	AD Stat.	p
SPX vs. ΔUST-10Y	32.24	$< 10^{-4}$
FTSE vs. ΔUK-10Y	∞	$< 10^{-4}$
DAX vs. ΔDE-10Y	41.78	$< 10^{-4}$
USD/JPY vs. DAX	78.50	$< 10^{-4}$

Anderson–Darling tests overwhelmingly reject normality for all residual series, confirming the presence of skewness and heavy tails. Financial time series are notorious for such properties, thus the results are unsurprising. However, bivariate normality is not a requirement of the theorem.

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